

4 CMOS Device Design

This chapter examines the key device design issues in a modern CMOS VLSI technology. It begins with an extensive review of the concept of MOSFET scaling. Two important CMOS device design parameters, threshold voltage and channel length, are then discussed in detail.

4.1 MOSFET Scaling

CMOS technology evolution in the past thirty years has followed the path of device scaling for achieving density, speed, and power improvements. MOSFET scaling was propelled by the rapid advancement of lithographic techniques for delineating fine lines of 1 μm width and below. In Section 3.2.1, we discussed that reducing the source-to-drain spacing, i.e., the channel length of a MOSFET, led to short-channel effects. For digital applications, **the most undesirable short-channel effect is a reduction in the gate threshold voltage at which the device turns on**, especially at high drain voltages. Full realization of the benefits of the new high-resolution lithographic techniques therefore requires the development of new device designs, technologies, and structures which can be optimized to keep short-channel effects under control at very small dimensions. Another necessary technological advancement for device scaling is in ion implantation, which not only allows the formation of very shallow source and drain regions but also is capable of accurately introducing a sharply profiled, low concentration of doping atoms for optimum channel profile design.

4.1.1

Constant-Field Scaling

In constant-field scaling (Dennard *et al.*, 1974), **it was proposed that one can keep short-channel effects under control by scaling down the vertical dimensions (gate insulator thickness, junction depth, etc.) along with the horizontal dimensions, while also proportionally decreasing the applied voltages and increasing the substrate**

objetivo: mantener invariante la ecuación de Poisson

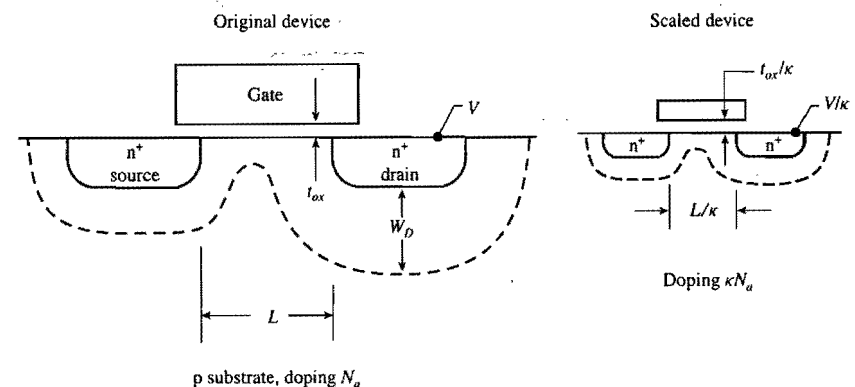


Figure 4.1. Principles of MOSFET constant-electric-field scaling. (After Dennard, 1986.)

doping concentration (decreasing the depletion width). This is shown schematically in Fig. 4.1. **The principle of constant-field scaling lies in scaling the device voltages and the device dimensions (both horizontal and vertical) by the same factor, κ (> 1), so that the electric field remains unchanged.** This assures that the reliability of the scaled device is not worse than that of the original device.

4.1.1.1

Diseno un canal largo y lo escala. No deberían aparecer efectos de canal corto

Rules for Constant-Field Scaling

Table 4.1 shows the scaling rules for various device parameters and circuit performance factors. The doping concentration must be increased by the scaling factor κ in order to keep Poisson's equation (3.66) invariant with respect to scaling. The maximum drain depletion width,

$$W_D = \sqrt{\frac{2\epsilon_{si}(\psi_{bi} + V_{dd})}{qN_a}}, \quad (4.1)$$

from Eq. (2.85) (with $V_{app} = -V_{dd}$) scales down approximately by κ provided that the power-supply voltage V_{dd} is much greater than the built-in potential ψ_{bi} . All capacitances (including wiring load) scale down by κ , since they are proportional to area and inversely proportional to thickness. The charge per device ($\sim C \times V$) scales down by κ^2 , while the inversion-layer charge density (per unit gate area), Q_i , remains unchanged after scaling. Since the electric field at any given point is unchanged, the carrier velocity ($v = \mu \mathcal{E}$) at any given point is also unchanged (the mobility is the same for the same vertical field). Therefore, any velocity saturation effects will be similar in the original and the scaled devices.

The drift current per MOSFET width, obtained by integrating the first term of the electron current density equation (2.54) over the inversion layer thickness, is

Cómo
escalat

Table 4.1 Scaling MOSFET device and circuit parameters

	MOSFET Device and Circuit Parameters	Multiplicative Factor ($\kappa > 1$)
Scaling assumptions	Device dimensions (t_{ox} , L , W , x_j)	$1/\kappa$
	Doping concentration (N_a , N_d)	κ
	Voltage (V)	$1/\kappa$
Derived scaling behavior of device parameters	Electric field ($\mathcal{E}$)	1
	Carrier velocity (v)	1
	Depletion-layer width (W_d)	$1/\kappa$
	Capacitance ($C = \epsilon A/t$)	$1/\kappa$
	Inversion-layer charge density (Q_i)	1
	Current, drift (I)	$1/\kappa$
	Channel resistance (R_{ch})	1
Derived scaling behavior of circuit parameters	Circuit delay time ($\tau \sim CV/I$)	$1/\kappa$
	Power dissipation per circuit ($P \sim VI$)	$1/\kappa^2$
	Power-delay product per circuit ($P\tau$)	$1/\kappa^3$
	Circuit density ($\propto 1/A$)	κ^2
	Power density (P/A)	1

$$\frac{I_{drift}}{W} = Q_i v = Q_i \mu \mathcal{E}, \quad (4.2)$$

and is unchanged with respect to scaling. This means that the drift current scales down by κ , consistent with the behavior of both the linear and the saturation MOSFET currents in Eq. (3.23) and Eq. (3.28). A key implicit assumption is that the threshold voltage also scales down by κ . Note that the velocity saturated current, Eq. (3.79), also scales the same way, since both v_{sat} and μ_{eff} are constants, independent of scaling. However, the diffusion current per unit MOSFET width, obtained by integrating the second term of the current density equation (2.54) and given by

$$\frac{I_{diff}}{W} = D_n \frac{dQ_i}{dx} = \mu_n \frac{kT}{q} \frac{dQ_i}{dx}, \quad (4.3)$$

scales up by κ , since dQ_i/dx is inversely proportional to the channel length. Therefore, the diffusion current does not scale down the same way as the drift current. This has significant implications in the nonscaling of MOSFET subthreshold currents, as will be discussed in Section 4.1.3.

4.1.1.2 Effect of Scaling on Circuit Parameters

With both the voltage and the current scaled down by the same factor, it follows that the **active channel resistance** [e.g., Eq. (3.102)] **of the scaled-down device remains unchanged**. It is further assumed that the parasitic resistance is either negligible or unchanged in scaling. The circuit delay, which is proportional to RC or CV/I , then scales

Beneficios de hacer el dispositivo más chico

down by κ . **This is the most important conclusion of constant-field scaling: once the device dimensions and the power-supply voltage are scaled down, the circuit speeds up by the same factor. Moreover, power dissipation per circuit, which is proportional to VI , is reduced by κ^2 .** Since the circuit density has increased by κ^2 , the power density, i.e., the active power per chip area, remains unchanged in the scaled-down device. This has important technological implications in that, in contrast to bipolar devices (Chapters 6, 7, and 8), packaging of the scaled CMOS devices does not require more elaborate heat-sinking. The power-delay product of the scaled CMOS circuit shows a dramatic improvement by a factor of κ^3 (Table 4.1).

4.1.1.3 Threshold Voltage

It was assumed earlier that the threshold voltage should be decreased by the scaling factor, κ , in proportion to the power-supply voltage. This is examined using the threshold equation (3.44) for a uniformly doped substrate:

$$V_t = V_{fb} + 2\psi_B + \frac{\sqrt{2\epsilon_{si}qN_a(2\psi_B - V_{bs})}}{C_{ox}} \quad (4.4)$$

where V_{bs} is the substrate bias voltage. In silicon technology, the material-related parameters (energy gap, work function, etc.) do not change with scaling; hence, in general, V_t does not scale. However, in a conventional process, n^+ -polysilicon gates are used for n-channel MOSFETs, and $V_{fb} = -E_g/2q - \psi_B$ from Eq. (2.209). It turns out that the first two terms on the RHS of Eq. (4.4) add up to approximately -0.15 V, which can be neglected. One can then argue (Dennard *et al.*, 1974) that by adjusting V_{bs} so that $2\psi_B - V_{bs}$ scales down by κ , the last term of Eq. (4.4), and therefore V_t , will also scale down by κ . However, at very early stages of the technology development, V_{bs} has been reduced to zero for most logic applications, though a reverse-biased body-source junction is still used for some dynamic memory array devices. Further reduction of the $2\psi_B - V_{bs}$ term with scaling would require a forward bias on the substrate. This is not commonly used in VLSI technologies, although it has been attempted in experimental devices (Sai-Halasz *et al.*, 1990). In practice, nonuniform doping profiles have been employed to tailor the threshold voltage of scaled devices, as will be discussed in Section 4.2.

p-channel MOSFETs with p^+ -polysilicon gates scale similarly to their counterparts. However, in buried-channel devices, e.g., when an n^+ -polysilicon gate is used for p-channel MOSFETs, the sum of the first two terms in Eq. (4.4) is nearly 1 V (magnitude) and therefore cannot be neglected. For this reason, it is difficult to scale buried-channel devices to low threshold voltages. More about threshold voltage design can be found in Section 4.2.

4.1.2 Generalized Scaling

Even though constant-field scaling provides a basic guideline to the design of scaled MOSFETs, the requirement of reducing the voltage by the same factor as the device

Table 4.2 CMOS VLSI technology generations

Feature size (μm)	Power-supply voltage (V)	Gate oxide thickness ($\text{\AA}$)	Oxide field (MV/cm)
2	5	350	1.4
1.2	5	250	2.0
0.8	5	180	2.8
0.5	3.3	120	2.8
0.35	3.3	100	3.3
0.25	2.5	70	3.6
0.1	1.5	30	5.0

physical dimension is too restrictive. Because of subthreshold nonscaling and reluctance to depart from the standardized voltage levels of the previous generation, the power-supply voltage was seldom scaled in proportion to channel length. Table 4.2 lists the supply voltage and device parameters of several generations of CMOS VLSI technology. It is clear that the oxide field has been increasing over the generations rather than staying constant. For device design purposes, therefore, it is necessary to develop a more general set of guidelines that allows the electric field to increase. In such a generalized scaling (Baccarani *et al.*, 1984), it is desired that both the vertical and the lateral electric fields change by the same multiplication factor so that the shape of the electric field pattern is preserved. This assures that 2-D effects, such as short-channel effects, do not become worse when scaling to a smaller dimension. Higher fields, however, do cause reliability concerns as mentioned in Section 2.5. Below $0.1 \mu\text{m}$ feature size, CMOS design space is severely constrained by power issues. Details of the performance-power tradeoff are discussed in Section 4.2.2.2 with the key parameters summarized in Fig. 4.7.

4.1.2.1 Rules for Generalized Scaling

If we assume that the electric field intensity changes by a factor of α , i.e., $\mathcal{E} \rightarrow \alpha \mathcal{E}$, while the device physical dimensions (both lateral and vertical) scale down by κ (>1) in generalized scaling, the potential or voltage will change by a factor equal to the ratio α/κ . If $\alpha = 1$, it reduces back to constant-field scaling. To keep Poisson's equation invariant under the transformation, $(x, y) \rightarrow (x, y)/\kappa$ and $\psi \rightarrow (\alpha/\kappa)\psi$ within the depletion region,

$$\frac{\partial^2(\alpha\psi/\kappa)}{\partial(x/\kappa)^2} + \frac{\partial^2(\alpha\psi/\kappa)}{\partial(y/\kappa)^2} = \frac{qN'_a}{\epsilon_{si}}, \quad (4.5)$$

N'_a should be scaled to $(\alpha/\kappa)N_a$. In other words, the doping concentration must be scaled up by an extra factor of α to control the depletion-region depth and thus avoid increased short-channel effects due to the higher electric field. Table 4.3 shows the generalized scaling rules of other device and circuit parameters.

Since the electric field intensity is usually increased in generalized scaling, the carrier velocity tends to increase as well. How much the velocity increases depends on how velocity-saturated the original device is. In the long-channel limit, carrier velocities are far from saturation and will increase by the same factor, α , as the electric field. The drift

Table 4.3 Generalized MOSFET scaling

	MOSFET device and circuit parameters	Multiplicative factor ($\kappa > 1$)	
Scaling assumptions	Device dimensions (t_{ox} , L , W , x_j)	$1/\kappa$	
	Doping concentration (N_a , N_d)	$\alpha \kappa$	
	Voltage (V)	α / κ	
Derived scaling behavior of device parameters	Electric field ($\mathcal{E}$)	α	
	Depletion-layer width (W_d)	$1/\kappa$	
	Capacitance ($C = \epsilon A/t$)	$1/\kappa$	
	Inversion-layer charge density (Q_i)	α	
Derived scaling behavior of circuit parameters		Long Ch.	Vel. Sat.
	Carrier velocity (v)	α	1
	Current, drift (I)	α^2 / κ	α / κ
	Circuit delay time ($\tau \sim CV/I$)	$1/\alpha \kappa$	$1/\kappa$
	Power dissipation per circuit ($P \sim VI$)	α^3 / κ^2	α^2 / κ^2
	Power-delay product per circuit ($P \tau$)		α^2 / κ^3
	Circuit density ($\propto 1/A$)		κ^2
	Power density (P/A)	α^3	α^2

current, which is proportional to $WQ_i v$, will then change by a factor of α^2/κ . This is consistent with the scaling behavior of long-channel currents, Eq. (3.23) and Eq. (3.28). On the other hand, if the original device is fully velocity-saturated, the carrier velocity cannot increase any more, in spite of the higher field in the scaled device. The current in this case will change only by a factor of α/κ , consistent with the velocity-saturated current, Eq. (3.81). The circuit delay scales down by a factor between κ and $\alpha\kappa$, depending on the degree of velocity saturation. The most serious issue with generalized scaling is the increase of the power density by a factor of α^2 to α^3 . This puts a great burden on VLSI packaging technology to dissipate the extra heat generated on the chip. The power-delay product is also a factor of α^2 higher than in constant-field scaling.

4.1.2.2 Constant-Voltage Scaling $\rightarrow$ *van der Plass*

Even though Poisson's equation within the depletion region is invariant under generalized scaling, the same is not true in the inversion layer when mobile charges are present. This is because mobile charge densities are exponential functions of potential which do not scale linearly with either physical dimensions or voltage. Furthermore, even in the depletion region, not all the boundary conditions scale consistently under generalized scaling. **This is due to the fact that the band bending at the source junction is given by the built-in potential (Appendix 9), which does not scale with voltage. Strictly speaking, the shape of the field pattern is preserved only if $\alpha = \kappa$, i.e., constant-voltage scaling.** Under constant-voltage scaling, the electric field scales up by κ and the doping concentration N_a scales up by κ^2 . The maximum gate depletion width (long-channel) [Eq. (3.68)],

$$W_{dm} = \sqrt{\frac{4\epsilon_{si} k T \ln(N_a/n_i)}{q^2 N_a}}, \quad (4.6)$$

then scales down by κ . Here $\ln(N_a/n_i)$ is a weak function of N_a and can be treated as a constant. This allows the short-channel V_t rolloff [Eq. (3.67)],

$$\Delta V_t = \frac{24t_{ox}}{W_{dm}} \left[\sqrt{\psi_{bi}(\psi_{bi} + V_{ds})} - a(2\psi_B) \right] e^{-\frac{\kappa L/2}{W_{dm} + 3t_{ox}}}, \quad (4.7)$$

to remain unchanged, as both t_{ox} and W_{dm} are scaled down by the same factor as the channel length L . Both the power-supply voltage and the threshold voltage [Eq. (4.4)],

$$V_t = V_{fb} + 2\psi_B + \frac{\sqrt{2\epsilon_{si}qN_a(2\psi_B - V_{bs})}}{C_{ox}}, \quad (4.8)$$

also remain unchanged. From Eq. (2.194), the inversion-layer charge per unit area is related to the electron concentration at the silicon surface, $n(0)$, by

$$Q_i = \sqrt{2\epsilon_{si}kTn(0)}. \quad (4.9)$$

Since Q_i scales up by κ in constant-voltage scaling, $n(0)$ scales up by κ^2 . Therefore, the mobile charge density scales the same way as the fixed charge density N_a . The inversion-layer thickness, being proportional to $Q_i/qn(0)$, scales down by κ just like other linear dimensions. The Debye length, $L_D = (\epsilon_{si}kT/q^2N_a)^{1/2}$, also scales down by κ under constant-voltage scaling.

Although constant-voltage scaling leaves the solution of Poisson's equation for the electrostatic potential unchanged except for a constant multiplicative factor in the electric field, it cannot be practiced without limit, since the power density increases by a factor of κ^2 to κ^3 . Higher fields also cause hot-electron and oxide reliability problems. In reality, CMOS technology evolution has followed mixed steps of constant-voltage and constant-field scaling, as is evident in Table 4.2.

4.1.3 Nonscaling Effects

4.1.3.1 Primary Nonscaling Factors

From the above discussions, it is clear that although constant-field scaling provides a basic framework for shrinking CMOS devices to gain higher density and speed without degrading reliability and power, there are several factors that scale neither with the physical dimensions nor with the operating voltage. **The primary reason for the nonscaling effects is that neither the thermal voltage kT/q nor the silicon bandgap E_g changes with scaling.** The former leads to subthreshold nonscaling; i.e., the threshold voltage cannot be scaled down like other parameters. The latter leads to nonscalability of the built-in potential, depletion-layer width, and short-channel effect.

From Eq. (3.40), the off current of a MOSFET is given by

$$I_{ds}(V_{gs} = 0, V_{ds} = V_{dd}) = \mu_{eff}C_{ox} \frac{W}{L} (m-1) \left(\frac{kT}{q} \right)^2 e^{-qV_t/mkT}. \quad (4.10)$$

Because of the exponential dependence, the threshold voltage cannot be scaled down significantly without causing a substantial increase in the off current. In fact, even if the

threshold voltage is held unchanged, the off current per device still increases by a factor of κ (from the C_{ox} factor) when the physical dimensions are scaled down by κ . This imposes a serious limitation on how low the threshold voltage can be, especially in dynamic circuits and random-access memories. The threshold voltage limitation in turn sets a lower limit on the power-supply voltage V_{dd} , since the circuit delay increases rapidly with the ratio V_t/V_{dd} , as will be discussed in Section 4.2.1.3.

Another nonscaling factor related to kT/q is the inversion-layer thickness, which is unchanged in constant-field scaling. Since the inversion-layer capacitance arising from the finite thickness is in series with the oxide capacitance, the total gate capacitance per unit area of the scaled device increases by a factor less than κ (Baccarani and Wordeman, 1983). This degrades the inversion charge density and therefore the current, especially at low gate voltages, as can be seen from Eq. (3.62).

Because both the junction built-in potential [Eq. (2.84)] and the maximum surface potential [Eq. (2.183)] are in the range of 0.6–1.0 V and do not change significantly with device scaling, the depletion-region widths, Eq. (4.1) and Eq. (4.6), do not scale quite as much as other linear dimensions. This results in worse short-channel effects in the scaled MOSFET, as is evident from Eq. (4.7). To compensate for these effects, the doping concentration must increase more than that suggested by constant-field scaling or generalized scaling.

4.1.3.2 Secondary Nonscaling Factors

Because of subthreshold nonscaling, the voltage level cannot be scaled down as much as the linear dimensions, and the electric field has increased as a result. This triggers several secondary nonscaling effects. First, in our discussions so far, it was implicitly assumed that carrier mobilities are constant, independent of scaling. However, as discussed in Section 3.1.5, the mobility decreases with increasing electric field:

$$\mu_{eff} \approx 32500 \mathcal{E}_{eff}^{-1/3}, \quad (4.11)$$

in units of $\text{cm}^2/\text{V}\cdot\text{s}$ for $\mathcal{E}_{eff} \leq 5 \times 10^5 \text{ V/cm}$. Beyond $\mathcal{E}_{eff} = 5 \times 10^5 \text{ V/cm}$, the mobility decreases even faster due to surface roughness scattering (Fig. 3.15). Since it is inevitable that the electric field increases with scaling, carrier mobilities are degraded in scaled MOSFETs. As a result, both the current and the delay improve less than the factors listed in Table 4.3 for generalized scaling. Furthermore, higher fields tend to push device operation more into the velocity-saturated regime. This means that the current gain and the delay improvement are closer to the velocity-saturated column of Table 4.3, and there is little to gain by operating at an even higher field or voltage.

The most serious problems associated with the higher field intensity are reliability and power. The power density increases by a factor of α^2 to α^3 as discussed before. Reliability problems arise from higher oxide fields, higher channel fields, and higher current densities. Even under the fully velocity-saturated condition, the current density increases by $\alpha\kappa$. This aggravates the problem of electromigration in aluminum lines, which is already becoming worse under constant-field scaling (Dennard *et al.*, 1974).

En resumen: esto es todo mentira y no sirve para una mierda

Higher fields also drive gate oxides closer to the breakdown condition, making it difficult to maintain oxide integrity. In fact, in order to curb the growing oxide field, the gate oxide thickness has been reduced less than the lateral device dimensions, e.g., the channel length, as is evident in Table 4.2. This means that the channel doping concentration must be increased more than called for in Table 4.3 to keep short-channel effects [Eq. (4.7)] under control. In other words, the maximum gate depletion width W_{dm} must be reduced more than the oxide thickness t_{ox} . This triggers another set of nonscaling effects, including the subthreshold slope $\propto m = 1 + (3t_{ox}/W_{dm})$, and the substrate sensitivity $dV_t/d(-V_{bs}) = m - 1$ [Eq. (3.45)]. These will be discussed in detail in Section 4.2.3.

4.1.3.3 Other Nonscaling Factors

In practice, there is yet another set of nonscaling factors encountered in CMOS technology evolution. One kind of nonscaling effect is related to the gate and source-drain doping levels. If not properly scaled up, they may lead to gate depletion and source-drain series resistance problems. From Eq. (2.213), polysilicon gate depletion contributes a capacitance $C_p = \epsilon_{eff} N_p / Q_g$ in series with the oxide capacitance C_{ox} . As C_{ox} increases by a factor of κ while Q_g remains unchanged in constant-field scaling, N_p must scale up by κ also to keep C_p in step with C_{ox} . In generalized scaling, N_p must scale up even more (by $\alpha\kappa$). In reality, this cannot be done because of limitations by solid solubility. The total gate capacitance then scales up by less than C_{ox} , leading to degradation of the inversion charge density and transconductance. Similarly, it is difficult to scale up the source-drain doping level and make the profile more abrupt while scaling down the junction depth. In practice, the source-drain series resistance has not been reduced in proportion to channel resistance, Eq. (3.102). This causes loss of current drive as the parasitic component becomes a more significant fraction of the total resistance in the scaled device.

Another class of nonscaling factors arises from process tolerances. The full benefit of scaling cannot be realized unless all process tolerances are reduced by the same factor as the device parameters. These include channel length tolerance, oxide thickness tolerance, threshold voltage tolerance, etc. **It is a key requirement and challenge in VLSI technology development to keep the tolerance to a constant percentage of the device parameter as the dimension is scaled down.** This could be a major factor in manufacturing costs as one tries to control a couple of hundred angstroms of channel length or a couple of atomic layers of gate oxide.

4.2

Threshold Voltage

This section focuses on a key design parameter in CMOS technology: threshold voltage. Although the threshold voltage was introduced in Chapter 3, the discussions there were restricted to the case of uniform doping. In this section, threshold-voltage requirements in terms of off- and on-currents are discussed, leading to the design of MOSFET channel profile with nonuniform body doping. The last two parts deal with the effects of quantum mechanics and dopant number fluctuations on threshold voltage.

4.2.1 Threshold-Voltage Requirement

4.2.1.1 Various Definitions of Threshold Voltage

First, we examine the various definitions of threshold voltage and the threshold-voltage requirement from a technology point of view. There are quite a number of different ways to define the threshold voltage of a MOSFET device. In Chapter 3 we followed the most commonly used definition [$\psi_s(\text{inv}) = 2\psi_B$] of V_t . The advantage of this definition lies in its popularity and ease of incorporation into analytical solutions. However, it is not directly measurable from experimental I - V characteristics (it can be determined from a split C - V measurement; see Exercise 2.6). In Section 3.1.6, we introduced the linearly extrapolated threshold voltage, V_{on} , determined by the intercept of a tangent through the maximum-slope (linear transconductance) point of the low-drain I_{ds} - V_{gs} curve. This is easily measured experimentally, but is about $3kT/q$ higher than the $2\psi_B$ threshold voltage, due to inversion-layer capacitance effects illustrated in Fig. 3.18.

Another commonly employed definition of threshold voltage is based on the subthreshold I_{ds} - V_{gs} characteristics, Eq. (3.40). For a given constant current level I_0 (say, 50 nA/ $\square$), one can define a threshold voltage V_t^{sub} such that $I_{ds}(V_{gs} = V_t^{sub}) = I_0(W/L)$. The advantages of such a threshold-voltage definition are twofold. First, it is easy to extract from hardware data and is therefore suitable for automated measurement of a large number of devices. Second, the device off current, $I_{off} = I_{ds}(V_{gs} = 0)$, can be directly calculated from I_0 , V_t^{sub} , and the subthreshold slope. In subsequent discussions, we will adhere to the $2\psi_B$ definition of V_t . In general, V_t depends on temperature (temperature coefficient), substrate bias (body-effect coefficient), channel length, and drain voltage (short-channel effect, or SCE).

4.2.1.2 Off-current and Standby Power

By definition, the off-current of a MOSFET is the source-to-drain subthreshold leakage current when the gate-to-source voltage is zero and the drain-to-source voltage is V_{dd} , the power supply voltage. From Eq. (3.40), the expression for the off-current with $V_{ds} = V_{dd} \gg kT/q$ is

$$I_{off} = I_{ds}|_{V_{gs}=0, V_{ds}=V_{dd}} = I_{ds,vt} e^{-qV_t/mkT}, \quad (4.12)$$

where

$$I_{ds,vt} = \mu_{eff} C_{ox} \frac{W}{L} (m-1) \left(\frac{kT}{q} \right)^2 \quad (4.13)$$

is defined as the source-to-drain current at threshold ($V_{gs} = V_t$, $V_{ds} = V_{dd}$). In the worst case, the source-drain voltage of the transistors in the off-state equals the power supply voltage V_{dd} . The standby power dissipation due to I_{off} is then $V_{dd} I_{off}$. For order-of-magnitude estimates, $V_{dd} \approx 1$ V. If it is desired that the standby power of a VLSI chip containing 10^8 transistors be no higher than 1 W, the off-current per transistor should be kept less than 10 nA.¹

¹ For a small fraction of transistors on the chip or for a larger standby power budget, higher off-current per transistor can be allowed.

Note that I_{ds, V_t} is rather insensitive to the temperature since $\mu_{eff} \propto T^{-3/2}$. However, it does depend on technology. For a 0.1 μm CMOS technology with $t_{ox} \approx 30 \text{ \AA}$, $\mu_{eff} \approx 350 \text{ cm}^2/\text{V}\cdot\text{s}$, $m \approx 1.3$, and $W/L = 10$, I_{ds, V_t} is approximately $1 \text{ }\mu\text{A}$ ($W = 1 \text{ }\mu\text{m}$). (Note that this number is for nMOSFETs. pMOSFET current is about $3\times$ lower due to the lower hole mobility. Also note that the extrapolated subthreshold current at the linearly extrapolated voltage V_{on} is about $10\times$ higher than this number, as discussed at the end of Subsection 3.1.6.4.) VLSI chips are usually specified for a worst-case temperature of 100°C where the off-current is much higher than that at room temperature because not only does V_t decrease with temperature, but the slope of the $\log(I_{ds})-V_{gs}$ curve also degrades in proportion to q/kT . Typically, the inverse slope of subthreshold current is 100 mV/decade at 100°C . For the factor $\exp(-qV_t/mkT)$ in Eq. (4.12) to deliver a two-orders-of-magnitude reduction from $I_{ds, V_t} = 1 \text{ }\mu\text{A}$ to $I_{off} = 10 \text{ nA}$, $V_t(100^\circ\text{C})$ needs to be at least 0.2 V . Because V_t has a negative temperature coefficient of $\approx -0.7 \text{ mV}/^\circ\text{C}$ (Section 3.1.4.2), this means $V_t(25^\circ\text{C}) \geq 0.25 \text{ V}$.²

The above figures are acceptable for CMOS logic technologies. In a dynamic memory technology (Dennard, 1984), however, the off-current requirement is much more stringent for the access transistor in the cell: on the order of $I_{off} \approx 10^{-13} - 10^{-14} \text{ A}$ (see Section 9.2.2). This means $V_t(100^\circ\text{C}) \geq 0.6 \text{ V}$ for a DRAM access device with $W = L = 0.1 \text{ }\mu\text{m}$. It should be noted that Eqs. (4.12) and (4.13) are analytical expressions derived under some simplifying approximations, e.g., long channel, uniform doping, etc. They are used here for order-of-magnitude estimates. More exact values of the off-current for a particular design should be obtained by numerical simulations.

Another consideration that may further limit how low the threshold voltage can be is the *burn-in* procedure. Burn-in is required in most VLSI technologies to remove early failures and ensure product reliability. It is usually carried out at elevated temperatures and over voltages to accelerate the degradation process. Both of these conditions further lower the threshold voltage and aggravate the leakage currents. Ideally, burn-in procedure should be designed such that it does not require a compromise on the device performance.

4.2.1.3 On-current and MOSFET Performance

While the lower bound of threshold voltage is set by standby power constraints, the upper bound is imposed by considerations of on-current and switching delay. The on-current of a MOSFET is defined in the saturation region as

$$I_{on} = I_{ds}|_{V_{gs}=V_{dd}, V_{ds}=V_{dd}} \quad (4.14)$$

Consider an nMOSFET initially in the off state with the source grounded and the drain charged to $V_{ds} = V_{dd}$ (e.g., in one of the CMOS inverter states in Fig. 5.2). If a gate voltage $V_{gs} = V_{dd}$ is applied to turn it on, the drain node will be discharged by the current I_{on} (initially) and the drain voltage will decrease at a rate given by

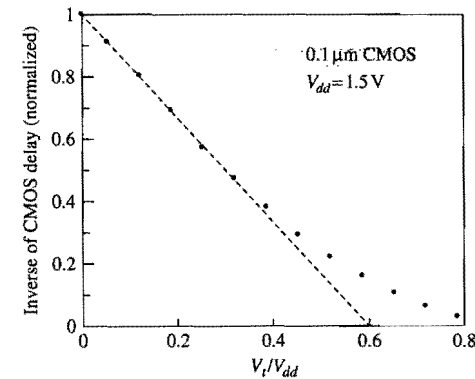


Figure 4.2. The reciprocal of CMOS delay in normalized units versus V_t/V_{dd} . The dots are from SPICE model simulations. The dashed line is a fitting proportional to $0.6 - V_t/V_{dd}$. Here V_t is defined as the gate voltage at which $I_{ds}(V_{ds} = V_{dd})$ equals that of Eq. (4.13). For a given linearly extrapolated, low-drain-bias threshold voltage V_{on} , a larger DIBL results in a lower V_t hence higher I_{off} and I_{on} .

$$C \frac{dV_{ds}}{dt} = -I_{on}, \quad (4.15)$$

where C is the total effective capacitance of the drain node. The switching delay for an incremental change of V_{ds} is then $-CdV_{ds}/I_{on} \propto 1/I_{on}$. It is evident from Chapter 3 that the lower the threshold voltage, the higher the current drive I_{on} , hence the faster the switching speed. From a CMOS performance point of view, it is desirable to have a threshold voltage as low as possible.

It will be discussed in Chapter 5 that because of the finite rise time of V_{gs} at the input, the current that goes into the discharge equation (4.15) is somewhat less than I_{on} . A circuit simulation model can be used to analyze the delay sensitivity to threshold voltage. Figure 4.2 shows a typical example of CMOS performance, defined as the reciprocal of CMOS delay, versus the normalized threshold voltage, V_t/V_{dd} . For $V_t/V_{dd} < 0.5$, the result can be fitted to an expression proportional to $0.6 - V_t/V_{dd}$. This indicates, for example, about 30% of the performance will be lost if V_t/V_{dd} is increased from 0.2 to 0.3. Because of such delay sensitivity, **the V_t/V_{dd} ratio is usually kept ≤ 0.25 for high performance CMOS circuits.**

4.2.1.4 I_{on} versus I_{off} Characteristics

Since the choice of threshold voltage hinges on the tradeoff between I_{off} and I_{on} , it is a common practice to plot I_{off} directly against I_{on} , thus skipping the ambiguous definition of threshold voltage. Figure 4.3 plots I_{ds} versus V_{gs} for a constant $V_{ds} = V_{dd}$ in both linear and logarithmic scales for the ease of reading I_{off} and I_{on} simultaneously. In essence, adjusting the threshold voltage of the device is equivalent to parallel shifting the $I_{ds}-V_{gs}$ curves horizontally along the V_{gs} -axis.

Note that for an incremental shift of $\Delta V_t > 0$, I_{off} decreases by a factor $\exp(q \Delta V_t / mkT)$ while I_{on} decreases by an amount $g_m \Delta V_t$, where $g_m = dI_{ds}/dV_{gs}$ is the saturation

² Lower threshold voltages are allowed in the scenario under footnote 1.

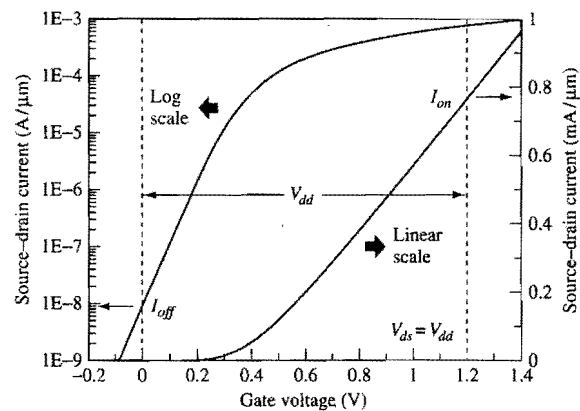


Figure 4.3. I_{ds} - V_{gs} characteristics in both linear and log scales; $V_{dd} = 1.2$ V in this example.

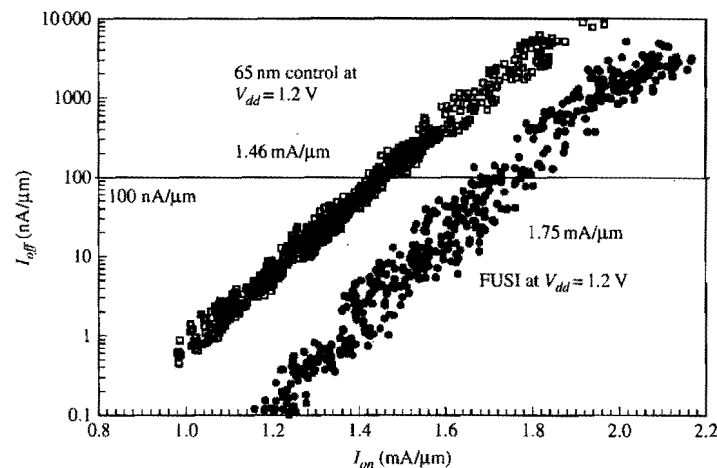


Figure 4.4. An experimental I_{off} - I_{on} plot for 65 nm nMOSFETs (Ranade *et al.*, 2005).

transconductance or the slope of the I_{ds} - V_{gs} curve at $V_{gs} = V_{dd}$. In this regard, the often cited I_{on}/I_{off} ratio is not a meaningful figure of merit because it changes constantly as ΔV_t is adjusted. In fact, to maximize the I_{on}/I_{off} ratio for a given V_{dd} , one would want to shift to as high a threshold voltage as possible so that the entire $0 \leq V_{gs} \leq V_{dd}$ range is in the subthreshold. That is not a desired mode of operation for high performance CMOS because then I_{on} would be so low that the delay is easily degraded by parasitic capacitances (Chapter 5).

An example of the recently published I_{on} - I_{off} characteristics for nMOSFETs is shown in Fig. 4.4 (Ranade *et al.*, 2005).

4.2.2 Channel Profile Design

In this section, we discuss the design of MOSFET doping profile that satisfies the threshold voltage and other device requirements. Parameters that come into play include the gate length, power supply voltage, and gate oxide thickness. The choice of gate work function is then addressed, leading to the channel profile requirements and trends over the CMOS technology generations.

4.2.2.1 CMOS Design Considerations

CMOS device design involves choosing a set of parameters that are coupled to a variety of circuit characteristics to be optimized. The choice of these device parameters is further subject to technology constraints and system compatibility requirements. Figure 4.5 shows a schematic diagram of the design process and the parameters involved. Because various circuit characteristics are interrelated through the device parameters, tradeoffs among them are often necessary. For example, reduction of W_{dm} improves short-channel effect, but degrades substrate sensitivity; thinner t_{ox} increases current drive, but causes reliability concerns, etc. There is no unique way of designing CMOS devices for a given technology generation. Nevertheless, we attempt here to give a general guideline of how these device parameters should be chosen.

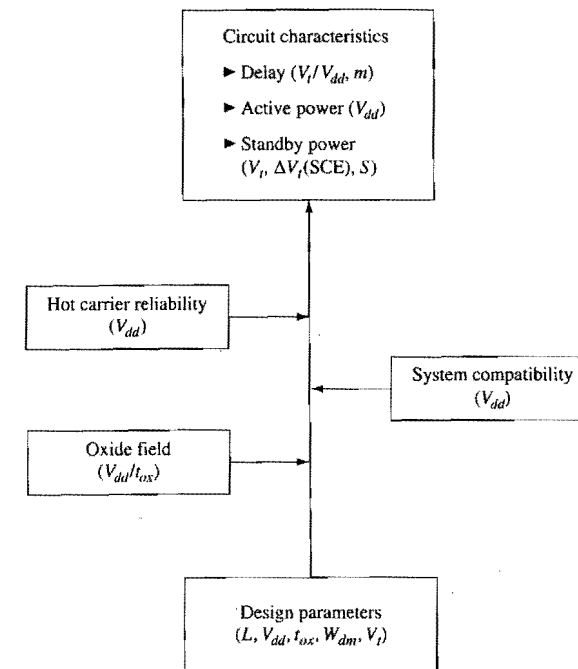


Figure 4.5. A CMOS design flowchart showing device parameters, technology constraints, and circuit objectives.

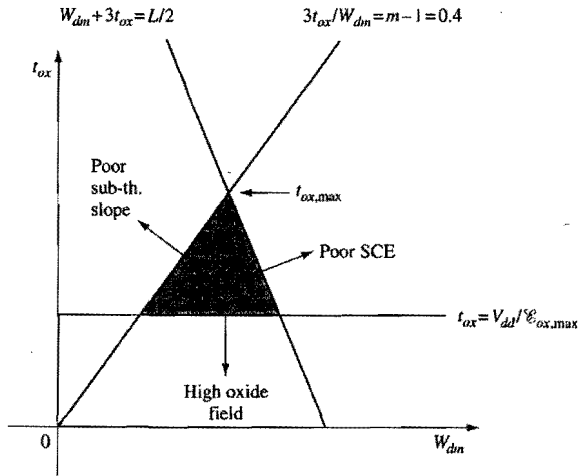


Figure 4.6. A t_{ox} - W_{dm} design plane. Some tradeoff among the various factors can be made within the parameter space bounded by SCE, body effect, and oxide field considerations.

Since threshold voltage plays a key role in determining both I_{off} and I_{on} , it is important to minimize the V_t tolerance, i.e., the spread between the high and low threshold voltages on the chip. The most dominant source of threshold voltage tolerances in a CMOS technology is from the short-channel effect. Channel length variations on a chip due to process imperfections give rise to threshold voltage variations. From Eq. (3.67) of Section 3.2.1.4, the short-channel V_t is lower than that of the long-channel by

$$\Delta V_t = \frac{24t_{ox}}{W_{dm}} \left[\sqrt{\psi_{bi}(\psi_{bi} + V_{ds})} - a(2\psi_B) \right] e^{-\frac{\pi L/2}{W_{dm} + 3t_{ox}}}, \quad (4.16)$$

where $a \approx 0.4$ and W_{dm} is the maximum depletion width at the threshold condition, $\psi_s = 2\psi_B$. The sensitivity of threshold voltage to channel length variations, $\delta V_t / \delta L$, is intimately tied to ΔV_t . Since $\psi_{bi} \approx 2\psi_B \approx 1$ V, and the worst case V_{ds} equals V_{dd} , the factor in the square bracket ranges between ≈ 1 and 2 V for $V_{dd} \approx 1$ V to 5 V. The factor in front of the square bracket, $24t_{ox}/W_{dm} = 8(m-1)$, is related to the factor $m = \Delta V_{gs} / \Delta \psi_s$ illustrated in Fig. 3.5. It was discussed in Sections 3.1.2.3, 3.1.3.3, and 3.1.4.1 that from saturation current, subthreshold slope, and substrate sensitivity considerations, m should not be too much greater than unity, e.g., $m \leq 1.4$. Because of the exponential factor in Eq. (4.16), ΔV_t is very sensitive to $L/(W_{dm} + 3t_{ox})$. A good choice is $L/(W_{dm} + 3t_{ox}) \geq 2$, which gives $\Delta V_t \leq 0.1$ V for $V_{dd} \approx 1$ V and $\Delta V_t \leq 0.2$ V for $V_{dd} = 5$ V, assuming a median value of $m = 1.3$.³

These considerations are captured in a plot of the t_{ox} - W_{dm} design plane in Fig. 4.6. The intercept of the two lines, $W_{dm} + 3t_{ox} = L/2$ and $3t_{ox}/W_{dm} = m - 1 = 0.4$, defines an upper

³ It can be shown from $\delta V_t / \delta L$ using Eq. (4.16) that this choice yields a δV_t spread equal to ΔV_t for a channel length tolerance $\delta L/L$ of $\pm 15\%$.

bound for the oxide thickness, $t_{ox,max} \approx L/20$. The lower limit of t_{ox} is imposed by technology constraints to $V_{dd}/\mathcal{E}_{ox,max}$, where $\mathcal{E}_{ox,max}$ is the maximum allowable oxide field from breakdown and reliability considerations. **For a given L and V_{dd} , the allowable parameter space in a t_{ox} - W_{dm} design plane is a triangular area bounded by SCE, oxide field, and subthreshold slope (also substrate sensitivity) requirements.**

In addition to the oxide field limitation, direct quantum mechanical tunneling (Fig. 2.62) also sets a lower limit to the thickness of gate oxide. Gate current density increases sharply as t_{ox} decreases below 2 nm. From Fig. 2.62, the gate tunneling current density for a 1 nm thick oxide biased at 1 V is 10^3 – 10^4 A/cm². Assume $L \approx 30$ nm, the gate current of an individual transistor (< 3 μ A/ μ m) is still small compared with the typical on currents (≈ 1 mA/ μ m) of the preceding stage so the switching delay of active transistors is hardly affected. But consider 10^8 transistors each with $W/L \approx 10$ and $L \approx 30$ nm, the total gate area per chip is of the order of 0.01 cm². The standby power dissipation of all the turned-on transistors⁴ in the chip has reached intolerable levels of 10–100 W. **Given the $t_{ox,max} \approx L/20$ criterion discussed above, the 1 nm t_{ox} limit translates into a channel length limit of ≈ 20 nm for SiO₂.**

If high- κ gate insulators become available, the scale length can be pushed to $\lambda \approx 2t_i$ for very high κ where t_i is the insulator thickness (Section 3.2.1.5). In that case, the minimum channel length can be extended to $2\lambda \approx 4t_i$, or ≈ 10 nm assuming a tunneling limited high- κ thickness of 2.5 nm. The last figure is thicker than that of SiO₂ because of the inherently lower barrier heights (< 3.1 eV of Fig. 2.29) of such materials.

4.2.2.2 Trends of Power Supply Voltage and Threshold Voltage

For a design window to exist in Fig. 4.6, it is required that $V_{dd}/\mathcal{E}_{ox,max} \leq t_{ox,max} \approx L/20$. This imposes an upper limit on the power supply voltage, namely,

$$V_{dd} \leq L\mathcal{E}_{ox,max}/20. \quad (4.17)$$

For $L = 1$ μ m CMOS technology, the gate oxides are relatively thick and $\mathcal{E}_{ox,max} \approx 3$ MV/cm. Equation (4.17) requires $V_{dd} \leq 15$ V. There is plenty of design room to choose the power supply and threshold voltages that satisfy both the off-current and the performance requirements discussed in Sections 4.2.1.2 and 4.2.1.3. For example, $V_{dd} = 5$ V and $V_t = 0.8$ – 1.0 V as shown in Fig. 4.7 in which the history and trends of power supply voltage, threshold voltage, and oxide thickness are plotted for CMOS logic technologies from 1.0 μ m to 0.02 μ m channel lengths (Taur *et al.*, 1995a). At shorter channel lengths, V_{dd} must be reduced. It becomes increasingly more difficult to satisfy both the performance and the off-current requirements. Fortunately, $\mathcal{E}_{ox,max}$ tends to increase for thinner oxides (see Section 2.5.6) as L is scaled down. This allows V_{dd} to scale at a slower rate than the channel length. Experimentally, $\mathcal{E}_{ox,max} \approx 6$ MV/cm for oxides thinner than 3 nm. Equation (4.17) then requires, e.g., that $V_{dd} \leq 1.5$ V for $L = 50$ nm CMOS technology. With such a low supply voltage, one often faces a tradeoff of circuit speed versus leakage current. Scaling down V_t causes I_{off} to increase exponentially. Even for the

⁴ The worst case gate leakage occurs with nMOSFETs biased at $V_{gs} = V_{dd}$ and $V_{ds} = 0$ (electrons tunnel from the inversion channel to the gate).

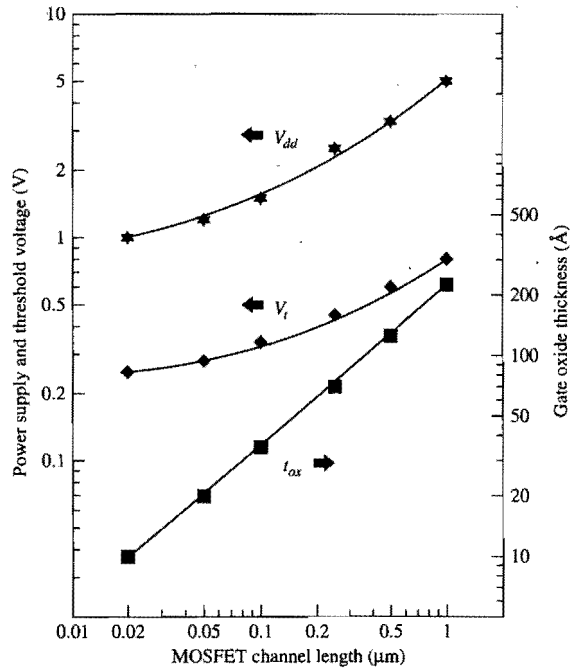


Figure 4.7. Trends of power-supply voltage, threshold voltage, and gate oxide thickness versus channel length for CMOS technologies from 1 μm to 0.02 μm . (After Taur *et al.*, 1995a.)

same V_t , I_{off} increases since I_{ds,V_t} of Eq. (4.13) increases as the devices are scaled down – a manifestation of subthreshold nonscalability. For this reason and for compatibility with the standardized power supply voltage of earlier generation systems, **the general trend is that V_{dd} has not been scaled down in proportion to L , and V_t has not been scaled down in proportion to V_{dd}** , as is evident in Fig. 4.7. At $L = 20 \text{ nm}$, $\mathcal{E}_{ox,max}$ is pushed to 10 MV/cm for operation at $V_{dd} = 1 \text{ V}$.

As a result of the non-scaled V_{dd} , not only does the field increase over the CMOS generations, the increasing power density (Table 4.3) also becomes more difficult to manage. It is discussed in Section 5.1.1 that the active or switching power of a CMOS circuit is given by

$$P_{ac} = CV_{dd}^2 f, \quad (4.18)$$

where C is the total equivalent capacitance being charged and discharged in a clock cycle, and f is the clock frequency. The power versus delay tradeoff can be represented conceptually in a V_{dd} – V_t design plane shown in Fig. 4.8 (Mii *et al.*, 1994). Higher performance, i.e., shorter delay, pushes for higher V_{dd} and lower V_t , which inevitably results in higher active power or higher standby power, or both. **Depending on the specific requirements of the application, CMOS technologies can be tailored to some extent by choosing an appropriate set of power supply and threshold voltages.** High

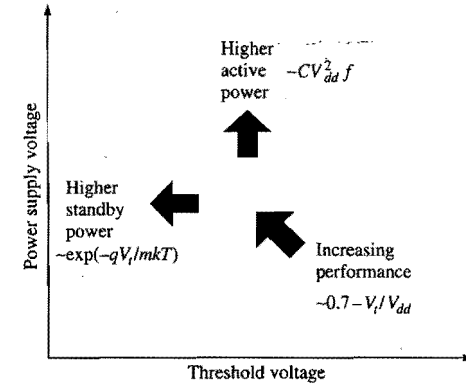


Figure 4.8. CMOS performance, active power, and standby power tradeoff in a V_{dd} – V_t design plane. The performance here is defined as the reciprocal of CMOS delay.

performance CMOS usually operates at the upper left-hand corner of the design space and pushes both power limits. Low power CMOS can operate at lower supply voltages and possibly at a higher threshold voltage if the standby power is of primary concern. It is a common practice in the state-of-the-art CMOS technologies to provide multiple threshold voltages on a chip to allow the design flexibility of using different types of devices for different functions, e.g., in memory and logic circuits. This comes, of course, at the expense of additional process complexity and cost.

4.2.2.3 Effect of Gate Work Function

To realize the threshold voltages desired from the above design considerations, it is important to use a gate material with the proper work function. Gate work function (ϕ_m) has a major impact on the threshold voltage of, e.g., nMOSFETs,

$$V_t = V_{fb} + 2\psi_B + \frac{-Q_d}{C_{ox}}, \quad (4.19)$$

since it sets the flatband voltage of the MOSFET,

$$V_{fb} = \phi_m - \phi_s = \phi_m - \left(\chi + \frac{E_g}{2q} + \psi_B \right). \quad (4.20)$$

For nMOSFETs, $2\psi_B \approx 1 \text{ V}$ and $Q_d < 0$, so V_t is easily larger than 1 V unless V_{fb} is substantially negative. To achieve the low threshold voltages required in Fig. 4.7, n⁺-polysilicon gates have been used for n-channel MOSFETs so that $V_{fb} = -E_g/2q - \psi_B$. This results in near cancellation of the first and the second terms of Eq. (4.19). V_t is then largely determined by the third term in proportion to the depletion charge density at the $2\psi_B$ condition. How the channel doping profile should be designed in order to achieve the desired depletion charge density and therefore V_t is discussed in the next subsection.

Before p⁺-polysilicon gates become technologically available, n⁺-polysilicon gates are used for pMOSFETs as well in 1 μm and 0.5 μm CMOS generations. This means V_{fb} is a small negative number ($V_{fb} = -E_g/2q + \psi_B$ for n-type silicon) and the first two terms of the

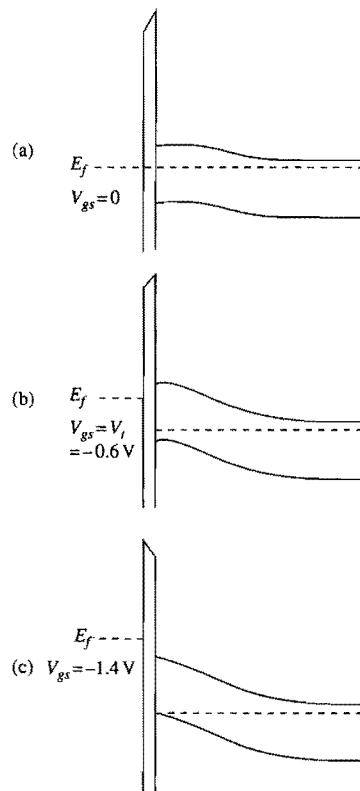


Figure 4.9. Band diagram of a buried-channel pMOSFET with n^+ -polysilicon gate. A shallow p-type layer is implanted at the surface to lower the magnitude of threshold voltage. The gate is biased (a) in subthreshold, (b) at threshold, and (c) beyond threshold. (After Taur *et al.*, 1985.)

V_t equation (the second term is $-2\psi_B \approx -1$ V for pMOS) add up to < -1 V for pMOSFETs. To make V_t less negative, the third term of the V_t equation needs to be positive, which means p-type doping for pMOSFETs or a counterdoped channel. Since the depletion charge density is negative, the surface field at threshold is such that holes are accelerated toward the substrate, and the channel for holes is formed at a potential minimum below the surface. Such devices are called *buried-channel* MOSFETs. Figure 4.9 shows the band diagrams of a buried-channel pMOSFET at several gate voltages both below and above the threshold. As the gate voltage becomes more negative than the threshold, the field changes sign and the channel moves to the surface. But the magnitude of the effective field is still lower than that of a conventional surface-channel device. **Although a buried-channel device has higher mobilities, its short-channel effect is inherently worse than that of a surface-channel device** (Nguyen and Plummer, 1981). This is because the counterdoping (especially boron) at the surface tends to diffuse deeper into the silicon during subsequent thermal cycles in the process. As the channel length and the power supply voltage are scaled down, a lower magnitude of threshold

voltage is required. It becomes increasingly more difficult to build a buried-channel device since higher counterdoping in the channel invariably leads to wider gate depletion widths and poorer short-channel effects. **For CMOS logic technologies of 0.25 μ m channel length and below, dual polysilicon gates (n^+ -polysilicon for nMOSFET and p^+ -polysilicon for pMOSFET) are used so that both types of devices are surface-channel devices** (Wong *et al.*, 1988).

Near the limits of CMOS scaling ($L \approx 10$ – 20 nm), the threshold voltages may become too high even with dual-polysilicon gates and extreme retrograde doping (Section 4.2.3.5). In principle, one way to further reduce the threshold magnitude is by counterdoping of the channel, as will be discussed in Section 4.2.3.6.

There have been numerous research explorations (e.g., Davari *et al.*, 1987) on using metal gates with a midgap work function. The benefits are high gate conductivity, absence of polysilicon depletion effects, and the simplicity of using a single gate material for both n- and pMOSFETs. Midgap work function gates exhibit symmetric flatband voltages: $V_{fb} = -\psi_B$ (p-type) for nMOSFETs and $V_{fb} = \psi_B$ (n-type) for pMOSFETs. The resulting threshold voltage magnitudes are in the range of 0.5–1.0 V [Eq. (4.19)]. This meets the V_t requirements for 1 μ m and 0.5 μ m CMOS technologies in Fig. 4.7. An added benefit is that it takes much less depletion charge [the third term in Eq. (4.19)] to achieve the same V_t magnitude with a midgap gate than with an n^+ -polysilicon gate for nMOSFETs. Less depletion charge means lower surface fields and therefore higher mobility. In reality, however, no midgap-work function gate material has been used in VLSI production because of technology issues such as compatibility of gate material with thin gate oxides. Gate conductivity requirement has been met with self-aligned silicide technology (Section 5.2.1.4). Once the CMOS technology is scaled to 0.25 μ m and below, V_t magnitudes < 0.5 V are needed (Fig. 4.7) which are difficult to achieve with a midgap work function gate.

4.2.2.4 Channel Profile Requirement and Trends

It was discussed above that with a n^+ -polysilicon gate for n-channel MOSFETs (and p^+ -poly for pMOSFETs), the first and the second terms of Eq. (4.19) essentially cancel out and V_t is largely determined by the depletion charge term. For a uniform channel doping, the maximum gate depletion width at the $2\psi_B$ condition,

$$W_{dm} = \sqrt{\frac{4\epsilon_{si}\psi_B}{qN_a}} \quad (4.21)$$

and the depletion charge term of the threshold voltage,

$$\frac{-Q_d}{C_{ox}} = \frac{qN_a W_{dm}}{C_{ox}} = \frac{\sqrt{4\epsilon_{si}qN_a\psi_B}}{C_{ox}}, \quad (4.22)$$

are coupled through the parameter N_a , and therefore cannot be varied independently (for a given t_{ox}). In Section 4.2.2.1, we discussed that in order to control the short-channel effect, $W_{dm} + 3t_{ox} = m W_{dm}$ should be on the order of $L/2$. The doping concentration that satisfies this requirement may not give the desired threshold voltage that satisfies the on- and off-current requirements.

For a given W_{dm} , it is necessary to employ nonuniform doping to adjust the depletion charge density to obtain the desired V_t . Nonuniform channel doping gives the device designer an additional degree of freedom to tailor the profile for meeting both the SCE and the threshold requirements. Such an optimization is made possible by the ion implantation technology.

Channel profile trends can be inferred by expressing the threshold voltage in the uniformly doped case as

$$V_t = V_{fb} + 2\psi_B + \frac{\sqrt{4\epsilon_{si}qN_a\psi_B}}{C_{ox}} = V_{fb} + 2\psi_B + 2(m-1)2\psi_B, \quad (4.23)$$

which does not scale much as neither m nor ψ_B changes significantly with channel length or doping. In fact, both m and ψ_B tend to increase slightly as the CMOS channel length scales down and higher doping is required. This is contrary to the downward trend of the V_t requirement depicted in Fig. 4.7. For example, for a typical $m \approx 1.3$, $V_t \approx 0.6$ V with n^+ -polysilicon gates. While this value happens to meet the V_t requirement for the $0.5 \mu\text{m}$ CMOS generation, it is too low for $1 \mu\text{m}$ CMOS and too high for CMOS generations $0.25 \mu\text{m}$ and below. It is shown in the next subsection that a high-low doping profile increases the depletion charge density for a given W_{dm} and therefore raises V_t over the uniformly doped value, whereas a low-high profile reduces the depletion charge and lowers V_t .

4.2.3 Nonuniform Doping

In this subsection, analytic expressions for the maximum depletion width and the threshold voltage are derived under nonuniform doping conditions. Specific results are given for both high-low and low-high doping profiles.

4.2.3.1 Integral Solution to Poisson's Equation

Mathematically the surface potential, electric field, and threshold voltage for the case of nonuniform channel doping can be solved using the depletion approximation. For a nonuniform p-type doping profile $N(x)$ in the same x -coordinate as defined in Fig. 2.35, the electric field is obtained by integrating Poisson's equation once (neglecting mobile carriers in the depletion region):

$$\mathcal{E}(x) = \frac{q}{\epsilon_{si}} \int_x^{W_d} N(x') dx', \quad (4.24)$$

where W_d is the depletion-layer width. Integrating again gives the surface potential,

$$\psi_s = \frac{q}{\epsilon_{si}} \int_0^{W_d} \int_x^{W_d} N(x') dx' dx \quad (4.25)$$

Using integration by parts, one can show that Eq. (4.25) is equivalent to (Brews, 1979)

$$\psi_s = \frac{q}{\epsilon_{si}} \int_0^{W_d} xN(x) dx. \quad (4.26)$$

The integral of $xN(x)$ equals the center of mass of $N(x)$ within $(0, W_d)$ times the integral of $N(x)$.

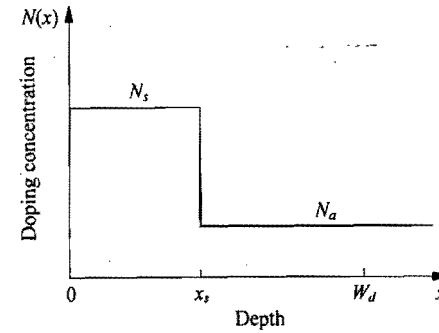


Figure 4.10. A schematic diagram showing the high-low step doping profile. $x=0$ denotes the silicon-oxide interface.

The maximum depletion-layer width (long-channel) W_{dm} is determined by the condition $\psi_s = 2\psi_B$ when $W_d = W_{dm}$. **The threshold voltage of a nonuniformly doped MOSFET is then determined by both the integral (depletion charge density) and the center of mass of $N(x)$ within $(0, W_{dm})$.**

4.2.3.2 A High-Low Step Profile

Consider the idealized step doping profile shown in Fig. 4.10 (Rideout *et al.*, 1975). It can be formed by making one or more low-dose, shallow implants into a uniformly doped substrate of concentration N_a . After drive-in, the implanted profile is approximated by a region of constant doping N_s that extends from the surface to a depth x_s . If the entire depletion region at the threshold condition is contained within x_s , the MOSFET can be considered as uniformly doped with a concentration N_s . The case of particular interest analyzed here is when the depletion width W_d exceeds x_s , so that part of the depletion region has a charge density N_s and part of it N_a . The integration in Eq. (4.26) can be easily carried out for this profile to yield the surface potential, or the band bending at the surface,

$$\psi_s = \frac{qN_s}{2\epsilon_{si}} x_s^2 + \frac{qN_a}{2\epsilon_{si}} (W_d^2 - x_s^2). \quad (4.27)$$

This equation can be solved for W_d as a function of ψ_s :

$$W_d = \sqrt{\frac{2\epsilon_{si}}{qN_a} \left(\psi_s - \frac{q(N_s - N_a)x_s^2}{2\epsilon_{si}} \right)}. \quad (4.28)$$

This is less than the depletion width in the uniformly doped (N_a) case for the same surface potential. The electric field at the surface is obtained by evaluating the integral in Eq. (4.24) with $x=0$:

$$\mathcal{E}_s = \frac{qN_s x_s}{\epsilon_{si}} + \frac{qN_a (W_d - x_s)}{\epsilon_{si}}. \quad (4.29)$$

From Gauss's law, the total depleted charge per unit area in silicon is given by

$$Q_s = -\epsilon_{si} \mathcal{E}_s = -qN_s x_s - qN_a(W_d - x_s), \quad (4.30)$$

as would be expected from Fig. 4.10. *The effect of the nonuniform surface doping is then to increase the depletion charge within $0 \leq x \leq x_s$ by $(N_s - N_a)x_s$ and, at the same time, reduce the depletion layer width as indicated by Eq. (4.28).*

Substituting Eq. (4.28) into Eq. (4.30) for Q_s , the gate voltage equation (3.14) becomes

$$V_{gs} = V_{fb} + \psi_s + \frac{1}{C_{ox}} \sqrt{2\epsilon_{si}qN_a \left(\psi_s - \frac{q(N_s - N_a)x_s^2}{2\epsilon_{si}} \right)} + \frac{q(N_s - N_a)x_s}{C_{ox}}. \quad (4.31)$$

By definition, the threshold voltage is the gate voltage at which $\psi_s = 2\psi_B$, i.e.,

$$V_t = V_{fb} + 2\psi_B + \frac{1}{C_{ox}} \sqrt{2\epsilon_{si}qN_a \left(2\psi_B - \frac{q(N_s - N_a)x_s^2}{2\epsilon_{si}} \right)} + \frac{q(N_s - N_a)x_s}{C_{ox}}. \quad (4.32)$$

The maximum depletion width (long-channel) at threshold is given by Eq. (4.28) with $\psi_s = 2\psi_B$:

$$W_{dm} = \sqrt{\frac{2\epsilon_{si}}{qN_a} \left(2\psi_B - \frac{q(N_s - N_a)x_s^2}{2\epsilon_{si}} \right)}. \quad (4.33)$$

There is some ambiguity as to whether $2\psi_B$ is defined in terms of N_s or N_a . We adopt the convention that $2\psi_B$ is defined in terms of the p-type concentration at the depletion-layer edge, i.e., $2\psi_B = (2kT/q) \ln(N_a/n_i)$. In fact, it makes very little difference which concentration we use, since $2\psi_B$ is a rather weak function of the doping concentration anyway.⁵ Further refinement of the threshold condition would require a numerical simulation of the specific profile.

In Section 3.1.3, we showed that the inverse subthreshold slope is given by $2.3mkT/q$ per decade where $m = dV_{gs}/d\psi_s$ at $\psi_s = 2\psi_B$. In the nonuniformly doped case, m can be evaluated from Eq. (4.31):

$$m \equiv \frac{dV_{gs}}{d\psi_s}(\psi_s = 2\psi_B) = 1 + \frac{\sqrt{2\epsilon_{si}qN_a}}{2C_{ox}} \left(2\psi_B - \frac{q(N_s - N_a)x_s^2}{2\epsilon_{si}} \right)^{-1/2}. \quad (4.34)$$

⁵ The " $2\psi_B$ " definition of threshold voltage is only a historical convention. Actually, the channel "turns on" when the surface potential is within 0.1 V (a few kT/q) of the conduction band edge of the n^+ source, regardless of the p-type body doping. In that respect, the approximation $2\psi_B \sim 1$ V is frequently used in the discussions.

It can be expressed in terms of W_{dm} using Eq. (4.33):

$$m = 1 + \frac{\epsilon_{si}/W_{dm}}{C_{ox}} = 1 + \frac{C_{dm}}{C_{ox}} = 1 + \frac{3t_{ox}}{W_{dm}}. \quad (4.35)$$

These expressions are consistent with Eq. (3.27) for a uniformly doped channel. This is to be expected from the basic concept of m in Fig. 3.5, which applies regardless of the doping specifics. Similarly, the threshold voltage in the presence of a substrate bias V_{bs} is given by Eq. (4.32) with the $2\psi_B$ term in the square root replaced by $2\psi_B - V_{bs}$. Using Eq. (4.33), one can show that the substrate sensitivity is

$$\frac{dV_t}{d(-V_{bs})} = \frac{\epsilon_{si}/W_{dm}}{C_{ox}} = \frac{C_{dm}}{C_{ox}} = m - 1. \quad (4.36)$$

Therefore, *all the previous expressions for the depletion capacitance, subthreshold slope, and body-effect coefficient in terms of W_{dm} for the uniformly doped case remain valid for the nonuniformly doped case.* The only difference is that the maximum depletion layer width W_{dm} in the high-low step doping case is given by Eq. (4.33) instead of Eq. (4.21).

4.2.3.3 Generalization to a Gaussian Profile

The results of the high-low step profile discussed above can be generalized to other profiles as well. As far as the threshold voltage and depletion width are concerned, the added doping density in Fig. 4.10, $N_s - N_a$ over $(0, x_s)$, is equivalent to the delta-function profile in Fig. 4.11(b) with an equivalent dose of

$$D_I = (N_s - N_a)x_s \quad (4.37)$$

centered at $x_c = x_s/2$. This is because both the integrals of $N(x)$ [Eq. (4.24)] and the center of mass of $N(x)$ [Eq. (4.26)] over $(0, W_{dm})$ are identical between the two profiles.

Similar arguments apply to a general Gaussian (or other symmetric) profile in Fig. 4.11(a) with a dopant distribution,

$$N(x) = \frac{D_I}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x - x_c)^2}{2\sigma^2}\right), \quad (4.38)$$

where σ is the implant straggle. The effect of such an implanted profile on threshold voltage and depletion-layer width is equivalent to that of the step doping profile discussed above, independent of σ . Substituting Eq. (4.37) and $x_c = x_s/2$ into the threshold voltage equation (4.32) yields

$$V_t = V_{fb} + 2\psi_B + \frac{1}{C_{ox}} \sqrt{2\epsilon_{si}qN_a \left(2\psi_B - \frac{qD_I x_c}{\epsilon_{si}} \right)} + \frac{qD_I}{C_{ox}}. \quad (4.39)$$

Similarly, the maximum depletion width, Eq. (4.33), becomes

$$W_{dm} = \sqrt{\frac{2\epsilon_{si}}{qN_a} \left(2\psi_B - \frac{qD_I x_c}{\epsilon_{si}} \right)}. \quad (4.40)$$

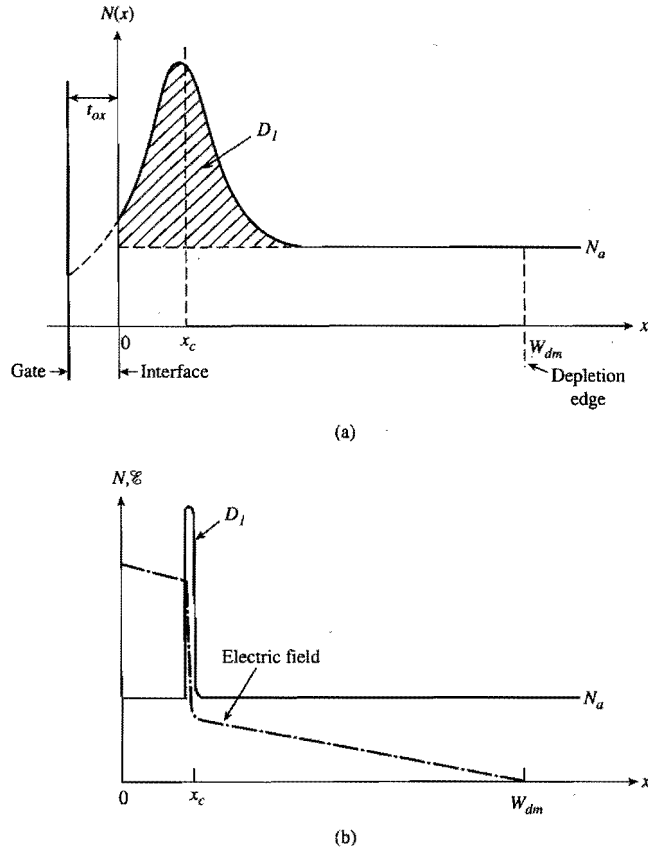


Figure 4.11. Schematic diagrams showing (a) an implanted Gaussian profile and (b) a delta-function profile equivalent to (a). The electric field is proportional to the area under the depleted charge $N(x)$ [Eq. (4.24)]. It has a step rise where the delta function doping is located. (After Brews, 1979.)

For a given implanted dose D_I , the resulting threshold voltage shift depends on the location of the implant, x_c . **For shallow surface implants, $x_c = 0$, there is no change in the depletion width. The V_t shift is simply given by qD_I/C_{ox} , as with a sheet of charge at the silicon-oxide interface.** All other device parameters, e.g., substrate sensitivity and subthreshold slope, remain unchanged. As x_c increases for a given dose, both the maximum depletion width and the V_t shift decrease. If x_c is not too large, one can always readjust the background doping N_a to a lower value N'_a to restore W_{dm} to its original value. The threshold voltage, in the meantime, is shifted by an amount less than the shallow implant case.

Although the above analysis on nonuniform doping assumes $N_s > N_a$, the results remain equally valid if $N_s < N_a$. Such a profile is referred to as the *retrograde channel doping*, discussed in the next subsection.

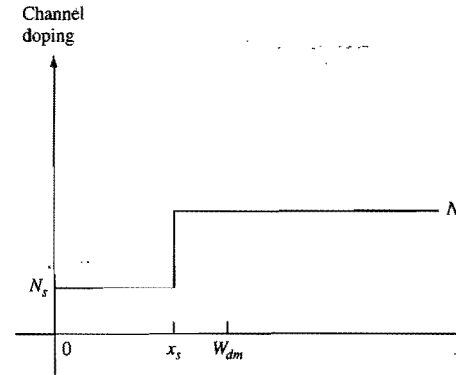


Figure 4.12. A schematic diagram showing the low-high (retrograde) step doping profile. $x=0$ denotes the silicon-oxide interface.

4.2.3.4 Retrograde (Low-High) Channel Profile

When the channel length is scaled to $0.25 \mu\text{m}$ and below, higher doping concentration is needed in the channel to reduce W_{dm} and control short-channel effects. If a uniform profile were used, the threshold voltage [Eq. (4.23)] would be too high even with dual polysilicon gates. The problem is further aggravated by quantum effects, which, as will be discussed in Section 4.2.4, can add another $0.1\text{--}0.2 \text{ V}$ to the threshold voltage because of the increasing fields (van Dort *et al.*, 1994).

To reduce the threshold voltage without significantly increasing the gate depletion width, a retrograde channel profile, i.e., a low-high doping profile as shown schematically in Fig. 4.12, is required (Sun *et al.*, 1987; Shahidi *et al.*, 1989). Such a profile is formed using higher-energy implants that peak below the surface. It is assumed that the maximum gate depletion width extends into the higher-doped region. All the equations in Section 4.2.3.2 remain valid for $N_s < N_a$. For simplicity, we assume an ideal retrograde channel profile for which $N_s = 0$. Equation (4.32) then becomes

$$V_t = V_{fb} + 2\psi_B + \frac{qN_a}{C_{ox}} \sqrt{\frac{4\epsilon_{si}\psi_B}{qN_a} + x_s^2} - \frac{qN_ax_s}{C_{ox}}. \quad (4.41)$$

Similarly, Eq. (4.33) gives the maximum depletion width,

$$W_{dm} = \sqrt{\frac{4\epsilon_{si}\psi_B}{qN_a} + x_s^2}. \quad (4.42)$$

The net effect of low-high doping is that the threshold voltage is reduced, but the depletion width has increased, just opposite to that of high-low doping. Note that Eq. (4.42) has the same form as Eq. (2.91) for a p-i-n diode discussed in Section 2.2.2. All other expressions, such as those for the subthreshold slope and the substrate sensitivity, in Section 4.2.3.2 apply with W_{dm} replaced by Eq. (4.42).

4.2.3.5 Extreme Retrograde Profile and Ground-Plane MOSFET

Two limiting cases are worth discussing. If $x_s \ll (4\epsilon_{si}\psi_B/qN_a)^{1/2}$, then W_{dm} remains essentially unchanged from the uniformly doped value [Eq. (4.42)], while V_t is lowered by a net amount equal to $qN_a x_s / C_{ox}$ [Eq. (4.41)]. In the other limit, N_a is sufficiently high that $x_s \gg (4\epsilon_{si}\psi_B/qN_a)^{1/2}$. In that case, $W_{dm} \approx x_s$, and the entire depletion region is undoped. All the depletion charge is concentrated at the edge of the depletion region. The square root term in Eq. (4.41) can be expanded into a power series to yield

$$V_t = V_{fb} + 2\psi_B + \frac{\epsilon_{si}(2\psi_B/x_s)}{C_{ox}}. \quad (4.43)$$

The last term stems from the depletion charge density in silicon, $\epsilon_{si}(2\psi_B/x_s)$, which can also be derived from Gauss's law by considering that the field in the undoped region is constant and equals $2\psi_B/x_s$ at threshold. Note that the work function difference that goes into V_{fb} is between the gate and the p^+ silicon at the edge of the depletion region. Using $m = 1 + 3t_{ox}/W_{dm} = 1 + 3t_{ox}/x_s$, one can write Eq. (4.43) as

$$V_t = V_{fb} + 2\psi_B + (m-1)2\psi_B. \quad (4.44)$$

Comparison with Eq. (4.23) shows that, with the extreme retrograde profile, the depletion charge (the third) term of V_t is reduced to half of the uniformly doped value. If there is a substrate bias V_{bs} present, the $2\psi_B$ factor in the last term of Eq. (4.44) is replaced by $(2\psi_B - V_{bs})$, i.e.,

$$V_t = V_{fb} + 2m\psi_B - (m-1)V_{bs}. \quad (4.45)$$

Since ψ_B is a weak function of N_a , the above results are independent of the exact value of N_a as long as it is high enough to satisfy $x_s \gg (4\epsilon_{si}\psi_B/qN_a)^{1/2}$. All the essential device characteristics, such as SCE (W_{dm}), subthreshold slope (m), and threshold voltage, are determined by the depth of the undoped layer, x_s . **The limiting case of retrograde channel profile therefore degenerates into a ground-plane MOSFET** (Yan et al., 1991). The band diagram and charge distribution of such a device at threshold condition are shown schematically in Fig. 4.13. Note that the field is constant (no curvature in potential) in the undoped region between the surface and x_s . There is an abrupt change of field at $x = x_s$, where a delta function of depletion charge (area = $2\epsilon_{si}\psi_B/x_s$) is located. Beyond x_s , the bands are essentially flat. It is desirable not to extend the p^+ region under the source and drain junctions, since that will increase the parasitic capacitance. The ideal channel doping profile is then that of a low-high-low type shown in Fig. 4.14, in which the narrow p^+ region is used only to confine the gate depletion width. Such a profile is also referred to as *pulse-shaped doping* or *delta doping* in the literature. The integrated dose of the p^+ region must be at least $2\epsilon_{si}\psi_B/qx_s$ to provide the gate depletion charge needed. It is advisable to use somewhat higher than the minimum dose to supply additional depletion charge to temper the source-drain fields in short-channel devices. However, too high a p^+ dose or concentration may result in band-to-band tunneling leakage between the source or drain and the substrate, as mentioned in Section 2.5.2.

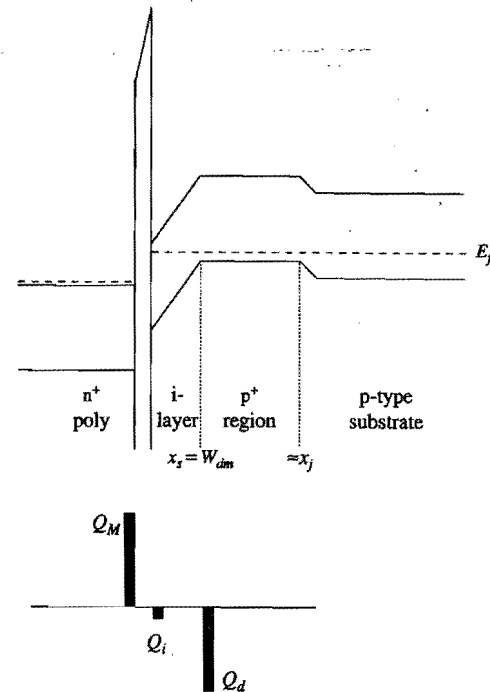


Figure 4.13. Band diagram and charge distribution of an extreme retrograde-doped or ground-plane nMOSFET at threshold condition.

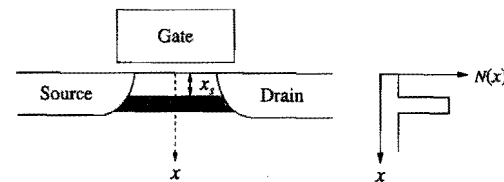


Figure 4.14. Schematic cross section of a low-high-low, or pulse-shaped, or delta-doped MOSFET. The doping concentration along the dashed line is depicted in the profile to the right. The highly doped region corresponds to the shaded area in the cross section.

4.2.3.6 Counter-Doped Channel

When CMOS devices are scaled to 20 nm channel lengths and below, the field is so high and the quantum effect so strong that even the extreme retrograde profile cannot deliver a $V_t \approx 0.2$ V with n^+ and p^+ silicon gates. Besides finding new gate materials with work functions outside of n^+ and p^+ silicon, further reduction of V_t can be accomplished, at least in principle, by either counterdoping the channel or forward biasing the substrate.

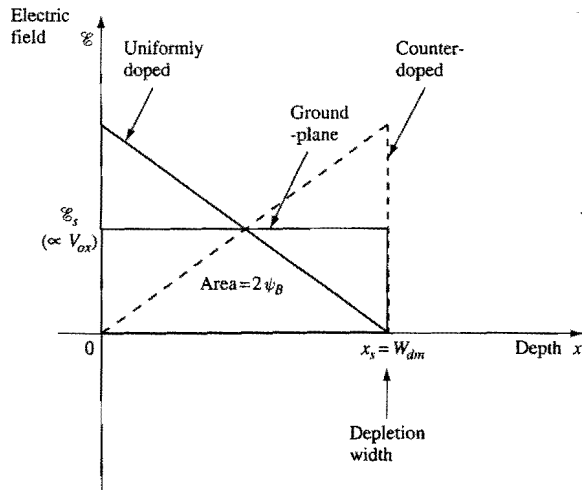


Figure 4.15. Graphical interpretation of uniformly doped, extreme retrograde or ground-plane, and counterdoped profiles. The band bending is given by the area under $\mathcal{E}(x)$ which equals $2\psi_B$ at threshold for all three cases.

A forward substrate bias also helps improve short-channel effects as it effectively reduces the built-in potential, ψ_{bi} in Eq. (3.67), between the source-drain and the p-type substrate. The flip side is that it causes source junction leakage, increases drain-to-substrate capacitance, and degrades subthreshold slope and body effect.

Instead of analyzing the counter-doped channel mathematically by letting $N_s < 0$ for the profile in Fig. 4.12, it is more instructive to give a graphical interpretation of the potential, field, and depletion charge by plotting the electric field versus depth as shown in Fig. 4.15. In the uniformly doped case, $\mathcal{E}(x)$ is a straight line with a negative slope whose magnitude is proportional to the substrate doping concentration N_a . The x -intercept gives the depletion layer width where $\mathcal{E} = 0$. The y -intercept gives the surface electric field $\mathcal{E}_s$ which from Gauss's law is proportional to the total depletion charge per unit area. Since $\mathcal{E} = -d\psi/dx$, the triangular area under $\mathcal{E}(x)$ equals the surface potential or the band bending ψ_s . As the gate voltage increases, both W_d and $\mathcal{E}_s$ increase and so does ψ_s until it reaches $2\psi_B$. At this point, surface inversion occurs and the depletion layer width has reached its maximum value. The depletion charge term of the threshold voltage, $V_{ox} = \epsilon_{si}\mathcal{E}_s/C_{ox}$, is proportional to the y -intercept or the surface field $\mathcal{E}_s$ when $\psi_s = 2\psi_B$.

In the extreme retrograde or the ground-plane case, $\mathcal{E}(x)$ is constant within the undoped region, $0 < x < x_s$, where there is no depletion charge. At the threshold condition, the shaded rectangular area for the ground-plane case is approximately the same as the triangular area under the uniformly doped $\mathcal{E}(x)$ since $2\psi_B$ is a rather weak function of N_a and can be considered as a constant for practical purposes. It is then clear that the depletion charge term of V_t or the y -intercept of the ground-plane case is half of that of the uniformly doped case – exactly as indicated by Eqs. (4.44) and (4.23).

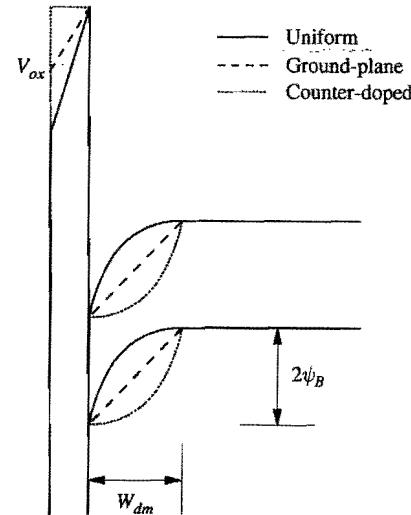


Figure 4.16. Band diagrams of uniformly doped, ground-plane (extreme retrograde), and counter-doped MOSFETs at threshold.

A specific case of the counter-doped channel is shown in Fig. 4.15. The slope $d\mathcal{E}/dx$ has the same magnitude as the uniformly doped case, but of the opposite polarity. Both the depletion width (x -intercept) and the band bending [area under $\mathcal{E}(x)$] are the same as the previous two cases. But the y -intercept ($\mathcal{E}_s$) is zero which means that the net charge in silicon is zero due to cancellation of the counter-doped charge with the depletion charge at the edge of the depletion region. This yields a very low V_t . Further counter-doping would result in $\mathcal{E}_s < 0$ or a buried channel MOSFET.

The band diagrams of these three doping cases at the threshold condition are further illustrated in Fig. 4.16. Both the depletion width and the band bending are kept the same for all three. But the surface fields (slopes) are very different, leading to dramatically different potential drops across the gate oxide.

4.2.3.7 Laterally Nonuniform Channel Doping

So far we have discussed nonuniform channel doping in the vertical direction. Another type of nonuniform doping used in very short-channel devices is in the lateral direction. For nMOSFETs, it is achieved by a medium-dose p-type implant carried out together with the n^+ source-drain implant after gate patterning. As shown in Fig. 4.17, the p-type doping peaks near the source and drain ends of the device but dips in the middle because of blocking of the implant by the gate. Such a self-aligned, laterally nonuniform channel doping is often referred to as *halo* or *pocket* implants (Ogura *et al.*, 1982). Figure 4.17 shows how halo works to counteract the short-channel effect, i.e., threshold rolloff toward the shorter devices within a spread of the channel length (or gate length). At the longer end of the spread shown in Fig. 4.17(a), the two p^+ pockets are farther apart than at the shorter end of the spread in Fig. 4.17(b). This creates a higher average p-type

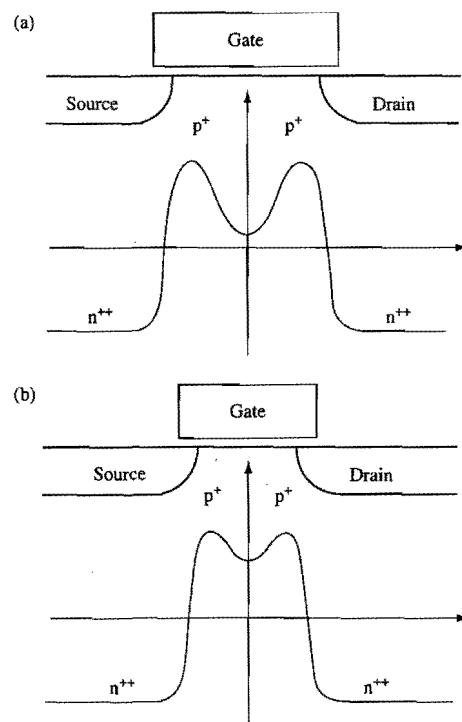


Figure 4.17. Laterally nonuniform halo doping in nMOSFETs. For a given design length on the mask, there is a spread of the actual gate lengths on the wafer. The longer end of the spread is shown in (a), the shorter in (b). The sketch below each cross section shows the schematic doping variation along a horizontal cut through the source and drain regions.

doping in the shorter device than in the longer device. Higher doping means higher threshold voltage. *So laterally nonuniform halo doping establishes a tendency for the threshold voltage to increase toward the shorter devices, which works to offset the short-channel effect in the opposite direction.* With an optimally designed 2-D nonuniform doping profile called the *superhalo*, it is possible in principle to counteract the short-channel effect and achieve nearly identical I_{on} and I_{off} in devices of different channel lengths within the process tolerances of a 25 nm MOSFET (Taur *et al.*, 1998).

4.2.4 Quantum Effect on Threshold Voltage

It was discussed in Section 2.3.2 that in the inversion layer of a MOSFET, carriers are confined in a potential well very close to the silicon surface. The well is formed by the oxide barrier (essentially infinite except for tunneling calculations) and the silicon conduction band, which bends down severely toward the surface due to the applied gate field. Because of the confinement of motion in the direction normal to the surface,

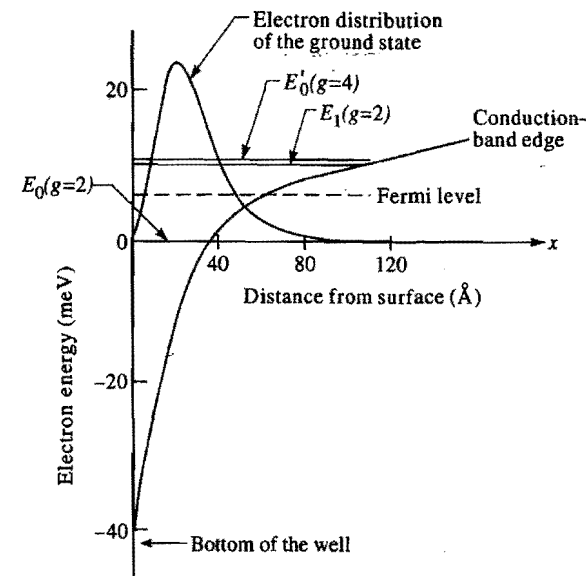


Figure 4.18. An example of quantum-mechanically calculated band bending and energy levels of inversion-layer electrons near the surface of an MOS device. The ground state is about 40 meV above the bottom of the conduction band at the surface. The dashed line indicates the Fermi level for 10^{12} electrons/cm² in the inversion layer. (After Stern and Howard, 1967.)

inversion-layer electrons must be treated quantum-mechanically as a 2-D gas (Stern and Howard, 1967), especially at high normal fields. Thus the energy levels of the electrons are grouped in discrete *subbands*, each of which corresponds to a quantized level for motion in the normal direction, with a continuum for motion in the plane parallel to the surface. An example of the quantum-mechanical energy levels and band bending is shown in Fig. 4.18. The electron concentration peaks below the silicon-oxide interface and goes to nearly zero at the interface, as dictated by the boundary condition of the electron wave function. This is in contrast to the classical model in which the electron concentration peaks at the surface, as shown in Fig. 4.19. Quantum-mechanical behavior of inversion-layer electrons affects MOSFET operation in two ways. *First, at high fields, threshold voltage becomes higher, since more band bending is required to populate the lowest subband at some energy above the bottom of the conduction band. Second, once the inversion layer forms below the surface, it takes a higher gate-voltage overdrive to produce a given level of inversion charge density.* In other words, the effective gate oxide thickness is slightly larger than the physical thickness. This reduces the transconductance and the current drive of a MOSFET.

4.2.4.1 Triangular Potential Approximation for the Subthreshold Region

A full solution of the silicon inversion layer involves numerically solving coupled Poisson's and Schrödinger's equations self-consistently (Stern and Howard, 1967).

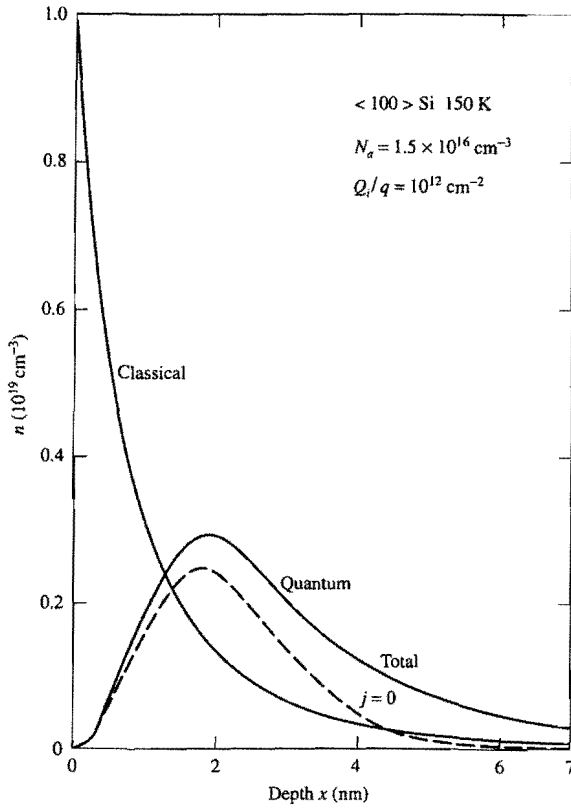


Figure 4.19. Classical and quantum-mechanical electron density versus depth for a $\langle 100 \rangle$ silicon inversion layer. The dashed curve shows the electron density distribution for the lowest subband. (After Stern, 1974.)

Under subthreshold conditions when the inversion charge density is low, band bending is solely determined by the depletion charge. It is then possible to decouple the two equations and obtain some insight into the quantum-mechanical (QM) effect on the threshold voltage. Since the inversion electrons are located in a narrow region close to the surface where the electric field is nearly constant ($\mathcal{E}_s$), it is a good approximation to consider the potential well as composed of an infinite oxide barrier for $x < 0$, and a triangular potential $V(x) = q\mathcal{E}_s x$ due to the depletion charge for $x > 0$. The Schrödinger equation is solved with the boundary conditions that the electron wave function goes to zero at $x = 0$ and at infinity. The solutions are Airy functions with eigenvalues E_j given by (Stern, 1972)

$$E_j = \left[\frac{3\hbar q \mathcal{E}_s}{4\sqrt{2m_x}} \left(j + \frac{3}{4} \right) \right]^{2/3}, \quad j = 0, 1, 2, \dots, \quad (4.46)$$

where $\hbar = 6.63 \times 10^{-34}$ J-s is Planck's constant, and m_x is the effective mass of electrons in the direction of confinement. Note that MKS units are used throughout this subsection (e.g., length must be in meters, not centimeters). The average distance from the surface for electrons in the j th subband is given by

$$x_j = \frac{2E_j}{3q\mathcal{E}_s}. \quad (4.47)$$

For silicon in the $\langle 100 \rangle$ direction, there are two groups of subbands, or *valleys*. The lower valley has a twofold degeneracy ($g = 2$) with $m_x = m_l = 0.92m_0$, where $m_0 = 9.1 \times 10^{-31}$ kg is the free-electron mass. These energy levels are designated as $E_0, E_1, E_2, \dots$. The higher valley has a fourfold degeneracy ($g' = 4$) with $m'_x = m_t = 0.19m_0$. The energy levels are designated as $E'_0, E'_1, E'_2, \dots$. Note that

$$E'_j = \left[\frac{3\hbar q \mathcal{E}_s}{4\sqrt{2m'_x}} \left(j + \frac{3}{4} \right) \right]^{2/3}, \quad j = 0, 1, 2, \dots \quad (4.48)$$

At room temperature, several subbands in both valleys are occupied near threshold, with a majority of the electrons in the lowest subband of energy E_0 above the bottom of the conduction band. From Appendix 12, the total inversion charge per unit area is expressed as (Stern and Howard, 1967)

$$Q_i^{QM} = \frac{4\pi q k T}{h^2} \left(g m_l \sum_j \ln(1 + e^{(E_f - E'_j - E_j)/kT}) + g' (m_t m_l)^{1/2} \sum_j \ln(1 + e^{(E_f - E'_j - E'_j)/kT}) \right), \quad (4.49)$$

where $m_t = 0.19m_0$ and $(m_l m_t)^{1/2} = 0.42m_0$ are the density-of-states effective masses of the two valleys, and $E_f - E'_j$ is the difference between the Fermi level and the bottom of the conduction band at the surface. It is shown in Appendix 12 that in the subthreshold region, Eq. (4.49) can be simplified to

$$Q_i^{QM} = \frac{4\pi q k T n_i^2}{h^2 N_c N_a} \left(2m_l \sum_j e^{-E_j/kT} + 4(m_l m_t)^{1/2} \sum_j e^{-E'_j/kT} \right) e^{q\psi_s/kT}, \quad (4.50)$$

where N_c is the effective density of states in the conduction band.

4.2.4.2 Threshold-Voltage Shift Due to Quantum Effect

When $\mathcal{E}_s < 10^4$ – 10^5 V/cm at room temperature, both the lowest energy level E_0 and the spacings between the subbands are comparable to or less than kT . A large number of subbands are occupied. It is shown in Appendix 12 that in this case, Q_i^{QM} is essentially the same as the classical inversion charge density per unit area given by Eq. (3.36) for the subthreshold region,

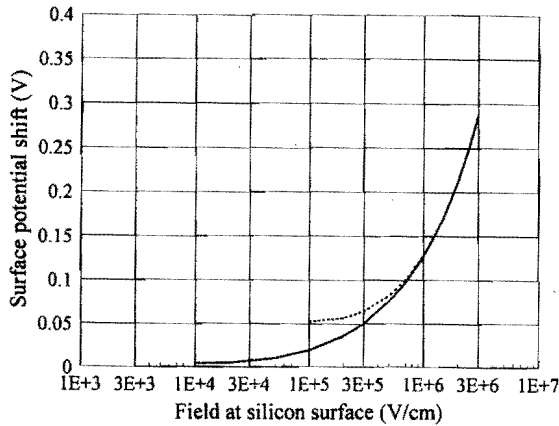


Figure 4.20. Additional band bending $\Delta\psi_s^{QM}$ (over the classical $2\psi_B$ value) required for reaching the threshold condition as a function of the surface electric field. The dotted curve is calculated by keeping only the lowest term (twofold degeneracy) in Eq. (4.50).

$$Q_i = \frac{kTn_i^2}{\mathcal{E}_s N_a} e^{q\psi_s/kT}. \quad (4.51)$$

(The expression has been generalized to cover nonuniformly doped cases where $\mathcal{E}_s$ is the electric field at the surface and N_a is the doping concentration at the edge of the depletion layer.) When $\mathcal{E}_s > 10^5$ V/cm, however, the subband spacings become greater than kT and Q_i^{QM} is significantly less than Q_i . The $Q_i^{QM} - \psi_s$ curve [Eq. (4.50)] exhibits a positive parallel shift with respect to the classical $Q_i - \psi_s$ curve [Eq. (4.51)] on a semilogarithmic scale, which means that additional band bending is required to achieve the same inversion charge per unit area as the classical value. The classical threshold condition, $\psi_s = 2\psi_B$, should therefore be modified to $\psi_s = 2\psi_B + \Delta\psi_s^{QM}$, where $Q_i^{QM}(\psi_s = 2\psi_B + \Delta\psi_s^{QM}) = Q_i(\psi_s = 2\psi_B)$. From this definition,

$$\Delta\psi_s^{QM} = \frac{kT}{q} \ln \left(\frac{Q_i(\psi_s = 0)}{Q_i^{QM}(\psi_s = 0)} \right) \quad (4.52)$$

can be evaluated from the preexponential factors in Eqs. (4.51) and (4.50). Figure 4.20 shows the calculated $\Delta\psi_s^{QM}$ as a function of $\mathcal{E}_s$. Beyond 10^6 V/cm, only the lowest subband is occupied by electrons and Eq. (4.52) becomes

$$\Delta\psi_s^{QM} \approx \frac{E_0}{q} - \frac{kT}{q} \ln \left(\frac{8\pi q m_i \mathcal{E}_s}{h^2 N_c} \right), \quad (4.53)$$

which is plotted as the dotted curve in Fig. 4.20. Knowing $\Delta\psi_s^{QM}$, one can easily calculate the threshold voltage shift due to the quantum effect:

$$\Delta V_t^{QM} = \frac{dV_{gs}}{d\psi_s} \Delta\psi_s^{QM} = m \Delta\psi_s^{QM}, \quad (4.54)$$

where $m = 1 + (3t_{ox}/W_{dm})$ as before.

As an example, consider a 50 nm MOSFET with a uniform doping of $N_a = 3 \times 10^{18}$ cm⁻³, which gives $W_{dm} = 20$ nm for control of short-channel effects. For this device, $\mathcal{E}_s \approx 10^6$ V/cm, so $\Delta\psi_s^{QM} = 0.13$ V from Fig. 4.20. If $m = 1.3$, then $\Delta V_t^{QM} = 0.17$ V, resulting in a much higher threshold voltage than the classical value. A retrograde doping profile not only reduces the depletion charge density (for a given W_{dm}) but also lowers the surface field hence ΔV_t^{QM} .

4.2.4.3 Quantum Effect on Inversion-Layer Depth

After strong inversion, the inversion charge density builds up rapidly and the triangular potential-well model is no longer valid. If the separation between the minimum energies of the lowest and the first excited subbands is large enough that only the lowest subband is populated, a variational approach leads to an approximate expression for the average distance of electrons from the surface (Stern, 1972):

$$x_{av}^{QM} = \left(\frac{9\epsilon_{si} h^2}{16\pi^2 m_x q Q^*} \right)^{1/3}, \quad (4.55)$$

where $Q^* = Q_d + \frac{11}{32} Q_i$ is a combination of the depletion and inversion charge per unit area in the channel. In general, the solution must be obtained numerically. Figure 4.21 shows a comparison of the classical and QM inversion-layer depths versus the effective normal field defined in Eq. (3.51) (Ohkura, 1990). The QM value is consistently larger than the classical value by about 10–12 Å for a wide range of channel doping (uniform) and effective fields. This degrades the inversion layer capacitance, $-dQ_i/d\psi_s = \epsilon_{si}/x_{av}$ (Section 3.1.6.2),⁶ and therefore the inversion charge component of the gate capacitance, $-dQ_i/dV_{gs} = \epsilon_{ox}/t_{inv}$. Effectively, the quantum-mechanical effect adds $\Delta t_{ox} = (\epsilon_{ox}/\epsilon_{si}) \Delta x_{av} = (x_{av}^{QM} - x_{av}^{CL})/3$ or about 3–4 Å to t_{inv} , causing lower current drive and transconductance in thin-oxide MOSFETs.

4.2.5 Discrete Dopant Effects on Threshold Voltage

As CMOS devices are scaled down, the number of dopant atoms in the depletion region of a minimum geometry device decreases. Due to the discreteness of atoms, there is a statistical random fluctuation of the number of dopants within a given volume around its average value. For example, in a uniformly doped $W=L=0.1$ -μm nMOSFET, if $N_a = 10^{18}$ cm⁻³ and $W_{dm} = 350$ Å, the average number of acceptor atoms in the depletion region is $N = N_a L W W_{dm} = 350$. The actual number fluctuates from device to device with a standard deviation $\sigma_N = \langle (\Delta N)^2 \rangle^{1/2} = N^{1/2} = 18.7$, which is a significant fraction of the average number N . Since the threshold voltage of a MOSFET depends on the charge of ionized dopants in the depletion region, this translates into a threshold-voltage fluctuation which could affect the operation of VLSI circuits.

⁶ Strictly speaking, the x_{av} in the capacitance is the center-of mass of the differential inversion charge responding to a differential change of ψ_s . Here we neglect the subtle difference.

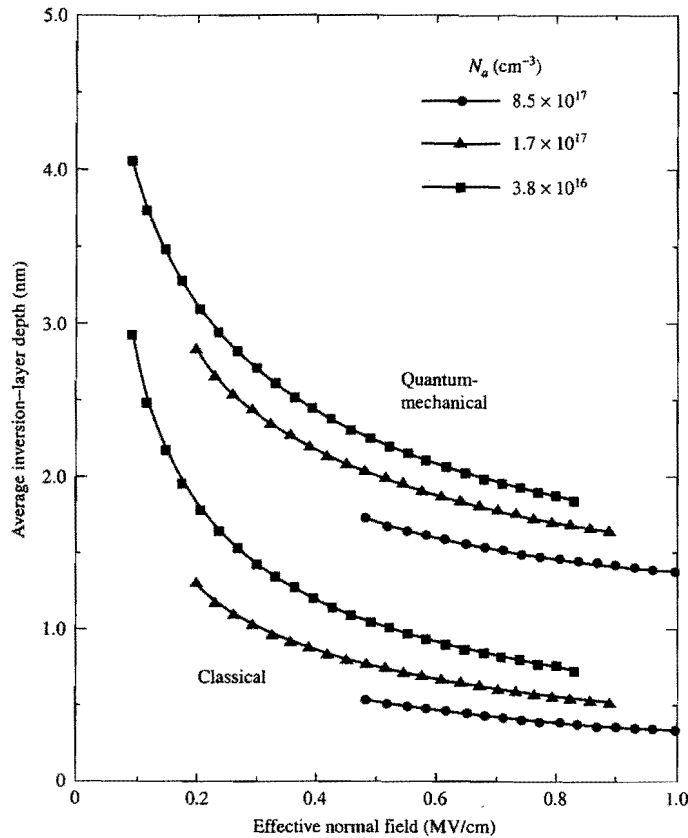


Figure 4.21. Calculated QM and classical inversion-layer depth versus effective normal field for several uniform doping concentrations. (After Ohkura, 1990.)

4.2.5.1 A Simple First-Order Model

To estimate the effect of depletion charge fluctuation on threshold voltage, we consider a small volume $dx dy dz$ at a point (x, y, z) in the depletion region of a uniformly doped (N_a) MOSFET. The x -axis is in the depth direction, the y -axis in the length direction, and the z -axis in the width direction. The average number of dopant atoms in this small volume is $N_a dx dy dz$. The actual number fluctuates around this value with a standard deviation of $\sigma_{dN} = (N_a dx dy dz)^{1/2}$. This fluctuation can be thought of as a small delta function of nonuniform doping (either positive or negative) at (x, y, z) superimposed on a uniformly doped background N_a . Here we focus on the linearly extrapolated threshold voltage V_{on} , as defined in Fig. 3.18. When there is a slight local nonuniformity of doping in either the channel-width or the channel-length direction, the first-order influence on the linear threshold voltage is through its effect on the total depletion charge integrated over the entire channel area (Nguyen, 1984). The effect of the above doping fluctuation on the

linear threshold voltage is then equivalent to that of a uniform (in W and L directions) delta-function implant of dose (number of ions per unit area) ΔD and depth x , where $\langle (\Delta D)^2 \rangle^{1/2} = \sigma_{dN}/WL = (N_a dx dy dz)^{1/2}/WL$. The threshold-voltage shift is obtained by substituting $D_f = \Delta D$ and $x_c = x$ in Eq. (4.39) and retaining only the first-order terms in ΔD :

$$\Delta V_{on} = \frac{q\Delta D}{C_{ox}} \left(1 - x \sqrt{\frac{qN_a}{2\epsilon_{si}(2\psi_B)}} \right) \approx \frac{q\Delta D}{C_{ox}} \left(1 - \frac{x}{W_{dm}} \right). \quad (4.56)$$

The last expression is quite general and is applicable to a nonuniformly doped background as well. It has its roots in Eq. (4.26). The mean square deviation (variance) of threshold voltage due to the depletion charge fluctuation in $dx dy dz$ is then

$$\langle \Delta V_{on}^2 \rangle_{x,y,z} = \frac{q^2 N_a}{C_{ox}^2 L^2 W^2} \left(1 - \frac{x}{W_{dm}} \right)^2 dx dy dz. \quad (4.57)$$

Since dopant number fluctuations at various points are completely random and uncorrelated, the total mean square fluctuation of the threshold voltage is obtained by integrating Eq. (4.57) over the entire depletion region:

$$\sigma_{V_{on}}^2 = \frac{q^2 N_a}{C_{ox}^2 L^2 W^2} \int_0^W \int_0^L \int_0^{W_{dm}} \left(1 - \frac{x}{W_{dm}} \right)^2 dx dy dz. \quad (4.58)$$

It is straightforward to carry out the integration and obtain

$$\sigma_{V_{on}} = \frac{q}{C_{ox}} \sqrt{\frac{N_a W_{dm}}{3LW}}. \quad (4.59)$$

In the above 0.1- μm example, $\sigma_{V_{on}} = 17.5 \text{ mV}$ if $t_{ox} = 35 \text{ \AA}$. This is small compared with the short-channel threshold rolloff in Section 4.2.2.1, but can be significant in minimum-geometry devices, for example, in an SRAM cell.

In the above analysis, it was assumed that the surface potential is uniform in both the length and the width directions of the device. In other words, all the lumpiness due to local fluctuations of the depletion charge is smoothed out and the surface potential depends only on the average (or total) depletion charge of the device. This assumption is not valid in the subthreshold region, where current injection is dominated by the highest potential barrier in the channel rather than by the average value (Nguyen, 1984). In general, the problem needs to be solved by 3-D numerical simulations (Wong and Taur, 1993). The results indicate that in addition to the threshold fluctuations of a similar magnitude to that expected from Eq. (4.59), there is also a negative shift of the average threshold voltage, especially in the subthreshold region. This is believed to be due to the inhomogeneity of surface potential resulting from the microscopic random distribution of discrete dopant atoms in the channel.

4.2.5.2 Discrete Dopant Effects in a Retrograde-Doped Channel

Threshold voltage fluctuations due to discrete dopants are greatly reduced in a retrograde-doped channel. Consider the profile in Fig. 4.12 with $N_s = 0$, i.e., the channel

is undoped within $0 < x < x_s$. The average threshold voltage and the maximum depletion width W_{dm} are given by Eq. (4.41) and Eq. (4.42), respectively. For a small volume of dopants at (x, y, z) where $x_s < x < W_{dm}$, Eq. (4.57) still holds. The x -integral in Eq. (4.58), however, is carried out from x_s to W_{dm} , which results in

$$\sigma_{V_{on}} = \frac{q}{C_{ox}} \sqrt{\frac{N_a W_{dm}}{3LW}} \left(1 - \frac{x_s}{W_{dm}}\right)^{3/2} \quad (4.60)$$

for a retrograde-doped channel. **In the extreme retrograde or ground-plane limit shown in Fig. 4.15, $x_s = W_{dm}$, and the threshold voltage fluctuation goes to zero.** This is also clear from Eq. (4.43), where the threshold voltage is essentially independent of N_a . Of course, the technological challenge is then to control the tolerance of the undoped-layer thickness x_s so that it does not introduce a different kind of threshold voltage variations.

In practice, retrograde channel doping reduces the threshold fluctuations due to discrete dopants, but does not eliminate them. For an optimally designed 25 nm MOSFET with superhalo (Taur *et al.*, 1998), a 3-D Monte-Carlo simulation has shown that the 1σ threshold voltage fluctuation due to discrete, random dopants is $10 \times W^{-1/2}$ mV where W is the device width in microns (Frank *et al.*, 1999). This is tolerable for logic devices with $W/L > 10$, but could be problematic for SRAM cell transistors which have minimum widths and require 6σ guard band for large arrays on a chip.

4.3 MOSFET Channel Length

Channel length is a key parameter in CMOS technology used for performance projection (circuit models), short-channel design, and model-hardware correlation. This section focuses on MOSFET channel length: its definition, extraction, and physical interpretation.

4.3.1 Various Definitions of Channel Length

A number of quantities, e.g., *mask length* (L_{mask}), *gate length* (L_{gate}), *metallurgical channel length* (L_{met}), and *effective channel length* (L_{eff}), have been used to describe the length of a MOSFET. Even though they are all related to each other, their relationships are strongly process-dependent.

Figure 4.22 shows schematically how various channel lengths are defined. L_{mask} is the design length on the polysilicon etch mask. It is reproduced on the wafer as L_{gate} through lithography and etching processes. Depending on the lithography and etching biases, L_{gate} can be either longer or shorter than L_{mask} . There are also process tolerances associated with L_{gate} . For the same L_{mask} design, L_{gate} may vary from chip to chip, wafer to wafer, and run to run. Although L_{gate} is an important parameter for process control and monitoring, there is no simple way of making a large number of measurements of it. Usually, L_{gate} is measured with a scanning electron microscope (SEM) and only sporadically across the wafer. There is also an uncertainty in the precise definition of L_{gate} when the polysilicon etch profile is not vertical, as to whether L_{gate} refers to the top or to the bottom dimension of the gate.

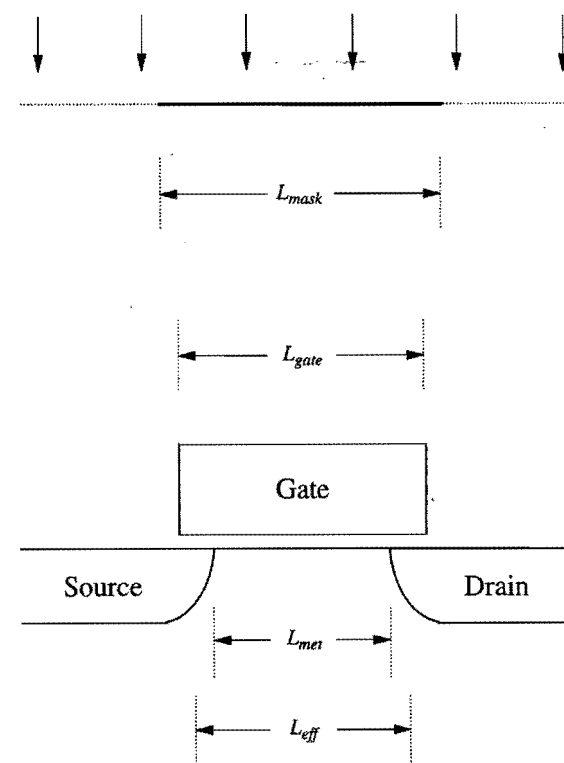


Figure 4.22. Schematic diagram showing the definitions of and relationship among the various notions of channel length. The physical interpretation of L_{eff} is examined in Section 4.3.3.

L_{met} is defined as the distance between the metallurgical junctions of the source and drain diffusions at the silicon surface. In a modern CMOS process, the source and drain regions are self-aligned to the polysilicon gate by performing the source-drain implant after gate patterning (Kerwin *et al.*, 1969). As a result, there is a close correlation between L_{met} and L_{gate} . Usually, L_{met} is shorter than L_{gate} by a certain amount due to the lateral implant straggle and the lateral source-drain diffusion in the process. Accurate physical measurement of L_{met} in actual hardware is very difficult. Normally, L_{met} is used only in 2-D models for short-channel device design. Even for that purpose, difficulties arise in defining L_{met} when dealing with a buried-channel device or a retrograde channel profile with zero surface doping, where there are no metallurgical junctions at the silicon surface.

The parameter L_{eff} is different from all other channel lengths discussed above in that it is defined through some electrical characteristics of the MOSFET device and is not a physical parameter. **Basically, L_{eff} is a measure of how much gate-controlled current a MOSFET delivers and is therefore most suitable for circuit models.** L_{eff} also allows for a large number of automated measurements, since it can be extracted from electrically measured terminal currents. The basis of the L_{eff} definition lies in the fact that the channel resistance of a MOSFET in the linear or low-drain bias region is proportional to the

channel length, as indicated by Eq. (3.102) (Dennard *et al.*, 1974). Further details of the definition and the extraction of L_{eff} are given in the next subsection.

For submicron CMOS technologies, it is important to distinguish among the various notions of channel length. The errors can be significant, since lithography and etching bias, junction depletion width, and lateral source-drain diffusions are all becoming an appreciable fraction of the channel length.

4.3.2 Extraction of the Effective Channel Length

As discussed in the last subsection, the effective channel length L_{eff} is defined by its proportionality to the linear or low-drain channel resistance. That is,

$$R_{ch} \equiv \frac{V_{ds}}{I_{ds}} = \frac{L_{eff}}{\mu_{eff} C_{inv} W (V_{gs} - V_{on} - m V_{ds}/2)} \quad (4.61)$$

from Eq. (3.102), where V_{on} is the linearly extrapolated threshold voltage and μ_{eff} is the effective mobility. C_{inv} contains the inversion-layer capacitance effect; μ_{eff} is a weak function of V_{gs} . For different L_{mask} , L_{eff} differs but is assumed to be related to L_{mask} by a constant channel length bias ΔL :

$$L_{eff} = L_{mask} - \Delta L. \quad (4.62)$$

All the lithography and etch biases as well as the lateral source-drain implant straggle and diffusion are lumped into ΔL . The assumption that the channel length bias is constant is a reasonable one when the channel length is not too short. However, ΔL can be linewidth-dependent when L_{mask} approaches the resolution limit of the lithography tool used in the process. This issue will be addressed later.

In the simplest scheme of channel-length extraction (Dennard *et al.*, 1974), R_{ch} is measured for a set of devices with different L_{mask} . Based on Eq. (4.61) and Eq. (4.62), a plot of R_{ch} for a given V_{gs} versus L_{mask} should yield a straight line whose intercept with the x-axis gives ΔL and therefore L_{eff} . In practice, however, two issues must be addressed for short-channel devices. The first one is the source-drain series resistance. The second one is the short-channel effect (SCE), which causes V_{on} in Eq. (4.61) to depend on L_{mask} .

4.3.2.1 Channel-Resistance Method

The effect of source-drain resistance is examined using the equivalent circuit in Fig. 4.23. A source resistance R_s and a drain resistance R_d are assumed to connect an *intrinsic* MOSFET to the external terminals where voltages V_{ds} and V_{gs} are applied. The internal voltages are V'_{ds} and V'_{gs} for the intrinsic MOSFET. One can write the following relations:

$$V'_{ds} = V_{ds} - (R_s + R_d)I_{ds} \quad (4.63)$$

and

$$V'_{gs} = V_{gs} - R_s I_{ds}. \quad (4.64)$$

As shown in Fig. 4.23, the intrinsic part of an actual device with parasitic resistance is equivalent to an intrinsic MOSFET with a grounded source, with V'_{gs} and V'_{ds} at the

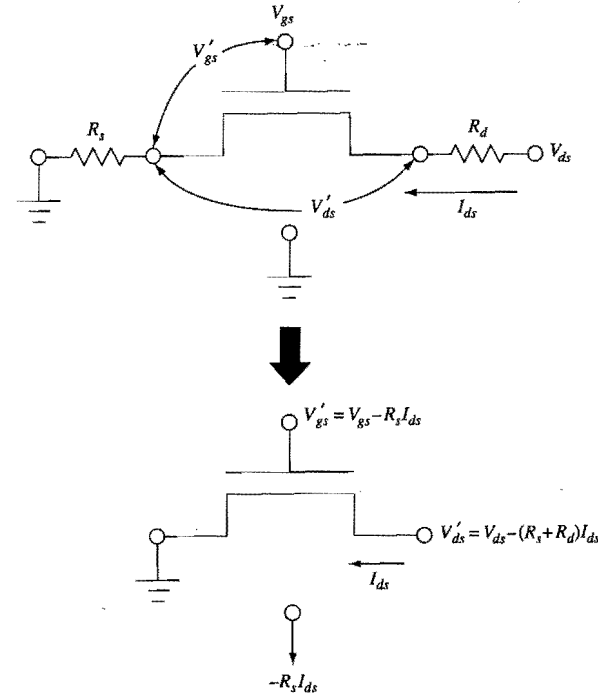


Figure 4.23. Equivalent circuit of MOSFET with source and drain series resistance. The intrinsic part of the top circuit is equivalent to the bottom circuit with redefined terminal voltages.

gate and the drain terminals, and with a reverse bias $-R_s I_{ds}$ on the substrate. Based on Eq. (4.61), but with redefined voltage symbols on the intrinsic nodes, the channel resistance of the intrinsic device is given by

$$R_{ch} \equiv \frac{V'_{ds}}{I_{ds}} = \frac{L_{eff}}{\mu_{eff} C_{inv} W (V'_{gs} - V'_{on} - m V'_{ds}/2)}, \quad (4.65)$$

where V'_{on} is the linear threshold voltage with the reverse bias on the substrate. It is related to the zero-substrate-bias threshold voltage V_{on} by

$$V'_{on} = V_{on} + (m - 1) R_s I_{ds}, \quad (4.66)$$

where $m - 1$ is the substrate sensitivity [Eq. (4.36)]. In a normal CMOS process, the source and drain regions are symmetrical, and therefore $R_s = R_d = R_{sd}/2$, where R_{sd} is the total source-drain parasitic resistance. Using Eqs. (4.62)–(4.66), one can write the externally measured total device resistance as

$$R_{tot} \equiv \frac{V_{ds}}{I_{ds}} = R_{sd} + R_{ch} = R_{sd} + \frac{L_{mask} - \Delta L}{\mu_{eff} C_{inv} W (V_{gs} - V_{on} - m V_{ds}/2)}. \quad (4.67)$$

Here all the internal voltages have been replaced by the voltages at the external terminals, since $V_{gs} - V_{on} - mV_{ds}/2 = V_{gs} - V_{on} - mV_{ds}/2$ from Eqs. (4.63), (4.64), and (4.66). Note that V_{on} is defined in terms of the intrinsic device, i.e., the threshold that would be obtained from linear extrapolation if there were no parasitic resistances.

For a set of devices with different L_{mask} but the same W , the parameters R_{sd} , ΔL , and C_{inv} are the same within process tolerances. It is also assumed that μ_{eff} does not change with channel length. The linear threshold voltage V_{on} , however, does depend on channel length because of short-channel effects. When comparing R_{tot} of devices with different L_{mask} , therefore, it is important to measure V_{on} for each device and adjust V_{gs} so that the gate overdrive $V_{gs} - V_{on}$ is the same from device to device. **A plot of R_{tot} (at small V_{ds}) versus L_{mask} for a given $V_{gs} - V_{on}$ will then yield a straight line that passes through the point $(\Delta L, R_{sd})$.** An example is shown in Fig. 4.24. The slope of the line depends on the

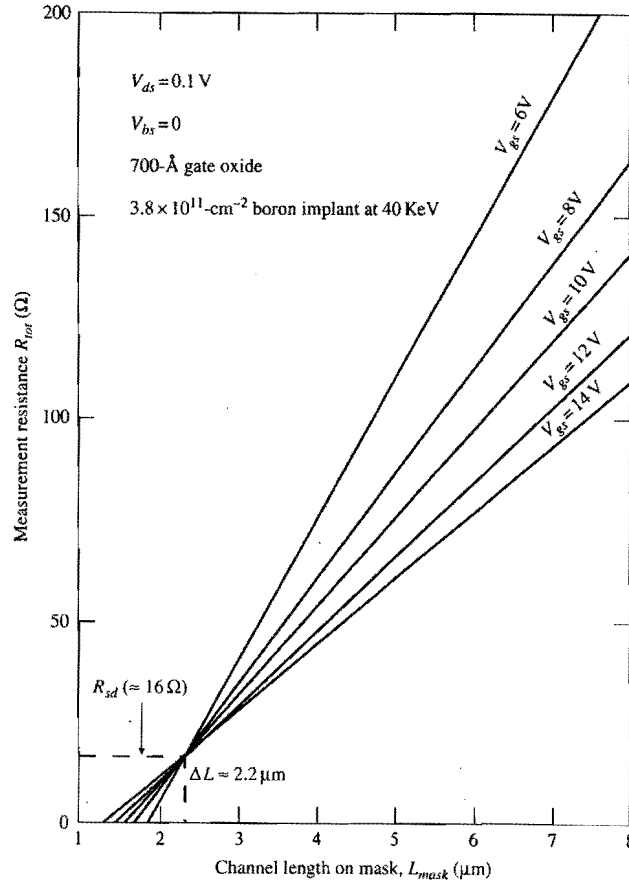


Figure 4.24. Measured R_{tot} at a low drain voltage versus L_{mask} for several different values of $V_{gs} - V_{on}$. The common intercept determines both ΔL and R_{sd} . (After Chern *et al.*, 1980.)

specific value of the gate overdrive. ΔL and R_{sd} are determined by the common intercept of several lines, each for a different $V_{gs} - V_{on}$ (Chern *et al.*, 1980).

4.3.2.2 Shift-and-Ratio Method

Despite the simplicity of the channel-resistance method described above, two main issues remain. First, it is not always straightforward to find the intrinsic V_{on} of short-channel devices. The presence of R_{sd} adds considerable difficulty in the usual linear extrapolation of V_{on} from the measured $I_{ds} - V_{gs}$ curve (Sun *et al.*, 1986). Typically, one tends to underestimate V_{on} in short-channel devices, as the degradation of I_{ds} by R_{sd} is more severe at higher currents. This introduces errors in channel-length extraction. The problem is further aggravated by a strong dependence of mobility on gate voltage, for example, in low-temperature and/or 0.1- μm MOSFETs. The second problem with the resistance method is that the R_{tot} -versus- L_{mask} lines for different gate overdrives may not intersect at a common point. Significant errors may result if only a limited number of $V_{gs} - V_{on}$ are investigated.

An improved channel-length extraction algorithm, called the *shift-and-ratio* (S&R) method, is able to circumvent the above problems (Taur *et al.*, 1992). This method is based on the same channel-resistance concept described above. It starts with a generalization of Eq. (4.67) to the form

$$R_{tot}^i(V_{gs}) = R_{sd} + L_{eff}^i f(V_{gs} - V_{on}^i), \quad (4.68)$$

where f is a general function of gate overdrive common to all the measured devices. The superscript i denotes the i th device, with an unknown effective channel length $L_{eff}^i = L_{mask}^i - \Delta L$ and linear threshold voltage V_{on}^i . The key assumption behind Eq. (4.68) is that the effective mobility μ_{eff} is a common function of $V_{gs} - V_{on}$ for all the measured devices. This is a reasonable assumption in view of Eq. (3.54) and Eq. (3.55).

The task is to calculate R_{sd} , L_{eff}^i , and V_{on}^i in Eq. (4.68) from the measured data on $R_{tot}^i(V_{gs})$. The S&R algorithm simplifies the procedure by differentiating Eq. (4.68) with respect to V_{gs} . Since the parasitic resistance R_{sd} is either independent or a weak function of V_{gs} , its derivative can be neglected:

$$S^i(V_{gs}) \equiv \frac{dR_{tot}^i}{dV_{gs}} = L_{eff}^i \frac{df(V_{gs} - V_{on}^i)}{dV_{gs}}. \quad (4.69)$$

Here df/dV_{gs} is also a general function of gate overdrive common to all the devices measured. An important benefit of working with the derivatives is that R_{sd} drops completely out of the picture, so it does not matter if R_{sd} varies from device to device as long as it is constant. An algorithm has been developed in the S&R method to take the ratio of $S^i(V_{gs})$ between two different devices – typically one long and one short, by shifting one along V_{gs} with respect to the other. The ratio becomes nearly constant, i.e., independent of V_{gs} , when the shift equals the difference between V_{on}^i of the two devices. L_{eff}^i is then determined from the ratio of the S functions at such a shift (Taur *et al.*, 1992).

4.3.3 Physical Meaning of Effective Channel Length

This subsection examines the physical meaning of L_{eff} extracted from electrically measured terminal currents. The effective channel length is defined through the linear channel resistance by Eq. (4.61). This equation is derived for long-channel devices and is not strictly valid for short-channel devices. By its definition, L_{eff} represents a measure of the effective gate-controlled resistance of the device and is not associated with any fixed physical quantity. When the channel profile is reasonably uniform and the source-drain doping is not too graded, L_{eff} is approximately equal to L_{met} (Laux, 1984). In general, however, one cannot take $L_{eff} = L_{met}$ for granted. The more graded (laterally) the source and drain profiles are, the longer L_{eff} is over L_{met} . This can be understood in terms of the spatial dependence of channel sheet resistivity discussed below.

4.3.3.1 Sheet Resistivity in Short-Channel Devices

Equation (4.61) implicitly assumes that the sheet resistivity, ρ_{ch} given by Eq. (3.103), is spatially uniform in both the MOSFET width and length directions. If the device is wide enough, ρ_{ch} can be considered uniform in that direction. However, the variation of ρ_{ch} in the length direction cannot be ignored in a short-channel device. From Eq. (3.8),

$$I_{ds} = -\mu_{eff} W Q_i(y) \frac{dV}{dy}, \quad (4.70)$$

where $V(y)$ is the quasi-Fermi level at a point y along the channel length direction. I_{ds} is a constant independent of y as required by current continuity. One can define a laterally varying sheet resistivity as

$$\rho_{ch}(y) = \frac{dV/dy}{I_{ds}/W} = \frac{1}{-\mu_{eff} Q_i(y)}. \quad (4.71)$$

Note that $Q_i < 0$ for nMOSFETs. This expression is valid as long as the current flow is largely parallel to the y -direction and the equipotential contours are perpendicular to the silicon surface. The total resistance is given by

$$\frac{V_{ds}}{I_{ds}} = \frac{1}{W} \int_{S-D} \rho_{ch}(y) dy, \quad (4.72)$$

where the integration is carried out from the heavily doped source region to the heavily doped drain region.

Figure 4.25 plots $\rho_{ch}(y)$ calculated from a 2-D device simulator versus distance at different gate voltages for both an abrupt and a graded source-drain (Taur *et al.*, 1995b). The area under each curve gives the total source-to-drain resistance at that gate voltage. In Fig. 4.25(a) for an infinitely abrupt (laterally) source-drain junction, the sheet resistivity is modulated by the gate voltage inside the (metallurgical) channel and independent of the gate voltage outside the (metallurgical) channel. However, in contrast to a long-channel device, $\rho_{ch}(y)$ is highly nonuniform, with a peak near the middle of the channel and decreasing toward the edges. This is due to SCEs from the source-drain fields, which help lower the potential barrier near the junctions and raise the local inversion

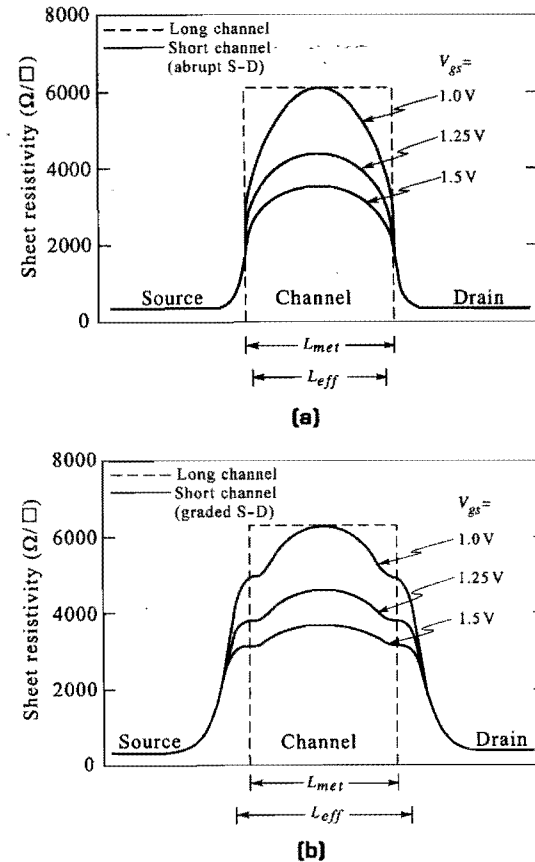


Figure 4.25. Simulated channel sheet resistivity at three different gate voltages versus distance from source to drain of an $L_{met} = 0.10\text{-}\mu\text{m}$ MOSFET. The curves in (a) are for an infinitely abrupt (laterally) source-drain which yields $L_{eff} = 0.091\text{ }\mu\text{m}$. The curves in (b) are for a graded (lateral straggle $\sigma_L = 165\text{ }\text{\AA}$) source-drain which yields $L_{eff} = 0.124\text{ }\mu\text{m}$. In both cases, the dashed lines represent the ideal, uniform-sheet resistivity of a scaled long-channel device. (After Taur *et al.*, 1995b.)

charge density there (Wordeman *et al.*, 1985). This effect is more pronounced at low gate voltages near threshold. The resulting L_{eff} extracted by the S&R method is slightly shorter than L_{met} .

Figure 4.25(b) shows similar plots for the same L_{met} , but with a finite lateral source-drain gradient. $\rho_{ch}(y)$ again is nonuniform inside the channel, being modulated by the gate voltage. In this case, however, a nonnegligible portion of the sheet resistivity outside the metallurgical channel is also gate-voltage-dependent. This is because of accumulation (Section 2.3.1) or gate modulation of the series resistance associated with the finite source-drain doping gradient. Since, according to the L_{eff} definition in Eq. (4.68), any part of the sheet resistivity that is gate-voltage dependent contributes to the effective

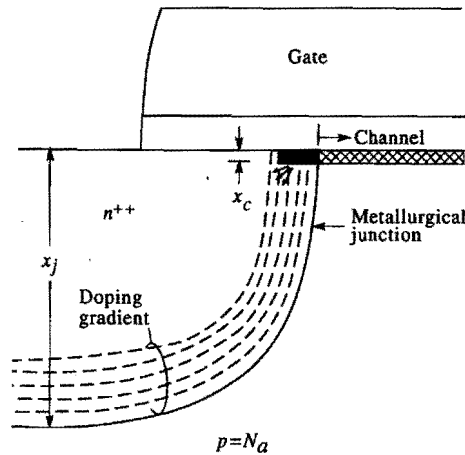


Figure 4.26. Schematic diagram showing doping distribution and current flow pattern near the end of the channel and the beginning of the source or drain. The dashed lines are contours of constant donor concentration, i.e., constant resistivity. The dark region represents the accumulation layer. (After Ng and Lynch, 1986.)

channel length, the extracted L_{eff} is substantially longer than L_{met} . At the same time, the extracted R_{sd} , which represents the constant part of the resistance in Eq. (4.68), only accounts for a portion of the series resistance outside the metallurgical channel.

4.3.3.2 Gate-Modulated Accumulation-Layer Resistance

Because of the finite lateral gradient of source-drain doping in practical devices, current injection from the surface inversion layer into the bulk source-drain region does not occur immediately at the metallurgical junction. When the gate voltage is high enough to turn on the MOSFET channel, an n^+ surface accumulation layer is also formed in the gate-to-source or -drain overlap region, as shown schematically in Fig. 4.26 (Ng and Lynch, 1986). Near the metallurgical junction and away from the surface, the donor concentration (also compensated by the p-type background) is low and the conductivity of the accumulation layer is higher than that of the bulk source-drain. As a result, current flow stays in the accumulation layer near the surface. This continues until the source-drain doping becomes high enough that the bulk conductance exceeds that of the accumulation layer. The point or region of current injection into the bulk depends on the lateral source-drain doping gradient. The more graded the profile is, the farther away the injection point is from the metallurgical junction.

The sheet resistivity of the accumulation layer can be estimated by applying Eq. (2.195) to the gate-to-source-drain overlap region:

$$V_{gs} = V_{fb} + \psi_s - \frac{Q_{ac}}{C_{ox}}, \quad (4.73)$$

where $Q_{ac} < 0$ is the accumulation charge (electrons) per unit area induced by the gate field, ψ_s is the band bending at the surface with respect to the bulk n-type region, and V_{fb} is the

flat-band voltage largely determined by the work-function difference between the gate electrode and the n-type silicon. For an n^+ -polysilicon-gated nMOSFET, $V_{fb} = -E_g/2q + \psi_B$, where ψ_B is given by Eq. (2.48) in terms of the local n-type doping concentration. The band bending in accumulation is approximately given by the distance between the n-type Fermi level and the conduction-band edge, i.e., $\psi_s \approx E_g/2q - \psi_B$. Therefore, V_{fb} and ψ_s in Eq. (4.73) nearly cancel each other and one obtains $V_{gs} \approx -Q_{ac}/C_{ox}$. The sheet resistivity of the accumulation layer is then

$$\rho_{ac} = \frac{1}{\mu_{ac}|Q_{ac}|} = \frac{1}{\mu_{ac}C_{ox}V_{gs}}, \quad (4.74)$$

where μ_{ac} is the average electron mobility in the accumulation layer (Sun and Plummer, 1980).

4.3.3.3 Interpretation of L_{eff} in Terms of Current Injection Points

The dependence of ρ_{ac} on V_{gs} in Eq. (4.74) is too similar to that of ρ_{ch} in Eq. (3.103) to allow separation of the accumulation-layer resistance from the channel resistance. **The region where the current flows predominantly in the accumulation layer is therefore considered as a part of L_{eff} .** The physical interpretation of L_{eff} in terms of injection points where the sheet resistivity of bulk source-drain equals that of the accumulation layer is consistent with 2-D device simulation results (Taur *et al.*, 1995b). For more graded (laterally) source-drain profiles, the injection points hence L_{eff} can be gate voltage dependent. At low gate overdrives, the injection point is closer to the metallurgical junction edge. As the gate voltage increases, the injection point moves out toward the more heavily doped source-drain region, resulting in a longer L_{eff} .

4.3.3.4 Implications for Short-Channel Effects

The fact that L_{eff} can be much longer than L_{met} has significant implications for the short-channel V_t rolloff curves. Figure 4.27 shows the low-drain threshold voltage rolloff versus L_{eff} for several different source-drain doping gradients. The abrupt doping profile has the best short-channel effect. As the lateral straggle σ_L increases, the short-channel effect becomes progressively worse. This can be understood from the above interpretation of L_{eff} . Current injection from the surface layer takes place at a certain source-drain doping concentration, e.g., 10^{19} cm^{-3} for nMOSFETs. For a given L_{eff} , the distance between the points where the doping concentration falls to 10^{19} cm^{-3} is fixed. The portion of the source-drain doping below 10^{19} cm^{-3} penetrates into the L_{eff} region from both ends. The more graded the source-drain profile is, the deeper such an n-type doping tail penetrates into the channel and compensates or reverses the p-type doping inside the channel. This is detrimental to the short-channel effect, as the edge regions become more easily depleted and inverted by the gate field (opposite to the halo effect). **It is therefore very important to reduce the width of the (laterally) graded source-drain region as the channel length is scaled down.**

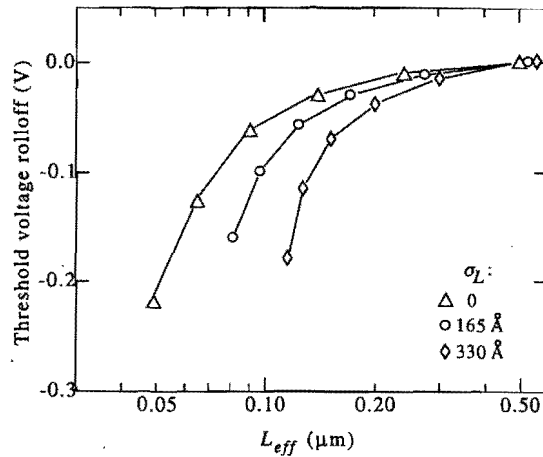


Figure 4.27. Simulated short-channel threshold rolloff versus L_{eff} for three different lateral source-drain doping gradients. On each curve, the points are for $L_{met} = 0.05, 0.07, 0.10, 0.15, 0.25$, and $0.50 \mu\text{m}$. (After Taur *et al.*, 1995b.)

4.3.4 Extraction of Channel Length by C - V Measurements

In an entirely different approach, another type of channel length has been extracted from the measured C - V data of a series of MOSFETs with different L_{mask} (Sheu and Ko, 1984). Capacitance measurements in general are more difficult to perform, as they require specially designed test sites. It is by no means straightforward to interpret the capacitively measured channel length and apply it to circuit models for current calculations.

The capacitive extraction of channel length is based on the fact that when a MOSFET is turned on, the intrinsic gate-to-channel capacitance is proportional to the channel length:

$$C_{gc} = C_{inv}WL_{cap}. \quad (4.75)$$

Here L_{cap} is the capacitively defined channel length, which may or may not be the same as L_{eff} or L_{met} . The gate-to-channel capacitance is usually measured in a split C - V setup that separates the majority-carrier response from the minority-carrier response, as shown in the inset of Fig. 4.28. The total measured capacitance consists of both the intrinsic gate-to-channel capacitance and a parasitic overlap capacitance from the gate to source-drain which is independent of channel length:

$$C_{tot} = C_{gc} + 2C_{ov} = C_{inv}WL_{cap} + 2C_{ov}. \quad (4.76)$$

Here C_{ov} is the overlap capacitance per gate edge (see Fig. 5.19). Typical examples of C_{tot} - V_{gs} curves are shown in Fig. 4.28 (Guo *et al.*, 1994). Using a large-area MOS capacitor, one can easily calibrate C_{inv} , taking all the polysilicon depletion and inversion-layer quantum effects into account. To find out L_{cap} , it is critical to determine what $2C_{ov}$ to subtract from the measured C_{tot} . In principle, $2C_{ov}$ in Eq. (4.76) is the parasitic

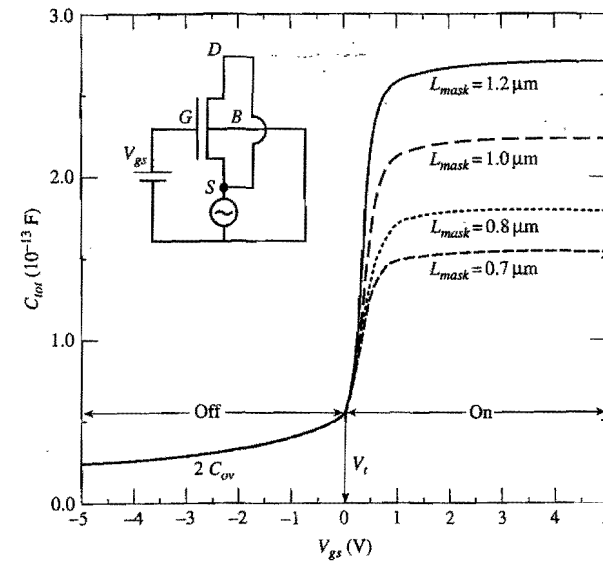


Figure 4.28. Example of measured capacitance from gate to source-drain versus gate voltage for MOSFETs of different mask lengths. The inset shows the split C - V measurement setup. (After Guo *et al.*, 1994.)

capacitance at a gate voltage when the MOSFET is on and C_{gc} is given by Eq. (4.75). In practice, $2C_{ov}$ cannot be separated from C_{gc} , since, unlike channel resistance, channel capacitance does not vary significantly with gate voltage once the device is turned on. What is usually done is to take $2C_{ov}$ as the measured capacitance when the MOSFET is off. However, from Fig. 4.28 it is clear that $2C_{ov}$ varies with the gate voltage (Oh *et al.*, 1990). There is no guarantee that $2C_{ov}$ in the off state is the same as $2C_{ov}$ in the on state. If $2C_{ov}$ is taken as the capacitance right below the threshold voltage, it will contain an unwanted inner-fringe term that is absent when the conducting channel is formed. If $2C_{ov}$ is taken at a negative gate voltage where the substrate is accumulated to eliminate the inner-fringe component, the lightly doped source-drain in the direct overlap region will be depleted (Sheu and Ko, 1984). Any such error in $2C_{ov}$ translates into a large error in L_{cap} when dealing with short-channel devices having small intrinsic capacitances.

A better interpretation of the capacitively extracted channel length is in terms of the gate length, L_{gate} (Fig. 4.22), since as the gate voltage varies, the same amount of charge per unit area is induced at the silicon surface whether it is in the inversion channel or in the source-drain overlap region under the gate. In other words, as far as the capacitance is concerned, the direct overlap length should be lumped into the channel length. This also circumvents the problem with the inner-fringe component mentioned above. One still needs to estimate the outer fringe capacitance and subtract it from the measured capacitance. But the outer fringe is smaller and can be estimated reasonably accurately using a simple formula (Section 5.2.2).

Exercises

- 4.1 Apply constant-field scaling rules to the long-channel currents [Eq. (3.23) for the linear region and Eq. (3.28) for the saturation region], and show that they behave as indicated in Table 4.1.
- 4.2 Apply constant-field scaling rules to the subthreshold current, Eq. (3.40), and show that instead of decreasing with scaling ($1/\kappa$), it actually increases with scaling (note that $V_{gs} < V_t$ in subthreshold). What if the temperature is also scaled down by the same factor ($T \rightarrow T/\kappa$)?
- 4.3 Apply constant-field scaling rules to the saturation current from the $n=1$ velocity saturation model [Eq. (3.79)] and the fully saturation-velocity limited current [Eq. (3.81)], and show that they behave as indicated in Table 4.1.
- 4.4 Apply generalized scaling rules to the saturation current from the $n=1$ velocity saturation model [Eq. (3.79)], and show that it behaves as indicated in Table 4.3 (between the two limits).
- 4.5 Consider an n-channel MOSFET with n^+ polysilicon gate (neglect poly depletion effect). The gate oxide is 7 nm thick, and the p-type body (or substrate) has a retrograde doping as shown in Fig. 4.12 with $N_s = 0$. Take $2\psi_B = 1$ V.
- Choose the values of x_s and N_a such that the maximum depletion width is $W_{dm} = 0.1 \mu\text{m}$ and the threshold voltage (at $2\psi_B$) is $V_t = 0.3$ V.
 - Following (a), what is the body effect coefficient, m , and the inverse slope of log subthreshold current versus gate voltage (long-channel device)?
 - Following (a), how short a channel length can the device be scaled to before short-channel effect becomes severe?
- 4.6 *Nonuniform V_t in the width direction.* A MOSFET is nonuniformly doped in the width direction. Part of the width (W_1) has a linear threshold voltage V_{on1} (defined in Fig. 3.18). The other part of the width (W_2) has a linear threshold voltage V_{on2} . Show that as far as the linear region characteristics are concerned, this device is equivalent to a uniform MOSFET of width $W_1 + W_2$ with a linear threshold voltage $V_{on} = (W_1 V_{on1} + W_2 V_{on2})/(W_1 + W_2)$. Ignore any fringing fields that may exist near the boundary between the two regions.
- 4.7 *Nonuniform V_t in the length direction.* A MOSFET is nonuniformly doped in the length direction. Part of the length (L_1) has a linear threshold voltage V_{on1} . The other part of the length (L_2) has a linear threshold voltage V_{on2} . Assume $V_{on1} \approx V_{on2}$, and consider only the first-order terms of $V_{on1} - V_{on2}$. Show that as far as the linear region characteristics are concerned, this device is equivalent to a uniform MOSFET of length $L_1 + L_2$ with a linear threshold voltage $V_{on} = (L_1 V_{on1} + L_2 V_{on2})/(L_1 + L_2)$. Ignore any fringing fields that may exist near the boundary between the two regions.
- 4.8 In the top equivalent circuit of Fig. 4.23, the source-drain current can be considered either as a function of the internal voltages: $I_{ds}(V'_{gs}, V'_{ds})$, or as a function of the external voltages: $I_{ds}(V_{gs}, V_{ds})$. The internal voltages are related to the external

voltages by Eqs. (4.63) and (4.64). Show that the transconductance of the intrinsic MOSFET can be expressed as

$$g'_m \equiv \left(\frac{\partial I_{ds}}{\partial V'_{gs}} \right)_{V'_{ds}} = \frac{g_m}{1 - g_m R_s - g_{ds}(R_s + R_d)},$$

where

$$g_m \equiv \left(\frac{\partial I_{ds}}{\partial V_{gs}} \right)_{V_{ds}}$$

is the extrinsic transconductance, and

$$g_{ds} \equiv \left(\frac{\partial I_{ds}}{\partial V_{ds}} \right)_{V_{gs}}$$

is the extrinsic output conductance.

- 4.9 Show that in the subthreshold region and when the drain bias is low, Eq. (3.12) leads to Eq. (4.51):

$$Q_i = \frac{kTn_i^2}{\mathcal{E}_s N_a} e^{q\psi_s/kT},$$

where ψ_s is the surface potential and $\mathcal{E}_s$ is the surface electric field. This equation is more general than Eq. (3.36) since it is valid for nonuniform (vertically) dopings with N_a being the p-type concentration at the edge of the depletion layer. (Note that the factor N_a merely reflects the fact in Fig. 2.32 that the band bending ψ_s is defined with respect to the bands of the neutral bulk region of doping N_a .)

- 4.10 In a short-channel device or in a nonuniformly doped (laterally) MOSFET, ψ_s may vary along the channel length direction from the source to drain. Generalize the expression in Exercise 4.9 and show that

$$\frac{V_{ds}}{I_{ds}} = \frac{1}{\mu_{eff} W} \int_0^L \frac{dy}{Q_i(y)} = \frac{N_a}{\mu_{eff} W k T n_i^2} \int_0^L \mathcal{E}_s(y) e^{-q\psi_s(y)/kT} dy$$

for the subthreshold region at low drain biases. Since $\mathcal{E}_s(y) \approx [V_{gs} - V_{fb} - \psi_s(y)]/3t_{ox}$ is not a strong function of ψ_s , the exponential factor dominates. This implies that the subthreshold current is controlled by the point of highest barrier (lowest ψ_s) in the channel. It also implies that the *channel length* factor entering the subthreshold current expression is different from the *effective channel length* defined by the linear region characteristics, Eq. (4.61).

- 4.11 Consider a uniformly doped nMOSFET of $N_a = 10^{18} \text{ cm}^{-3}$ biased at the threshold condition. Calculate the first three quantum mechanical energy levels for inversion electrons in the lower valley with an effective mass of $0.92m_0$ where m_0 is the free electron mass. Express the answers in eV.
- 4.12 For an nMOSFET with $t_{ox} = 10 \text{ nm}$ and a uniform p-type doping of 10^{17} cm^{-3} , the gate is n^+ polysilicon doped to 10^{20} cm^{-3} . Estimate the depletion layer width in the polysilicon gate at a gate voltage of 3 V.